
Federated Learning for Data-Locality-Preserving Distributed Space Weather Monitoring: A Solar-Flare Prediction Component for Grid-Resilience Pipelines

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September 2026

Abstract — The vulnerability of modern power grids to geomagnetically induced currents (GICs), triggered by M- and X-class solar flares, presents a critical and recurring threat to bulk electric infrastructure; the 1989 Québec system collapse remains the canonical demonstration of how a single geomagnetic storm can disable an entire provincial grid. Effective early warning therefore requires predictive models trained on magnetogram-derived feature histories, and the framework developed here targets a deployment scenario in which participating institutions impose data-locality or data-sharing constraints that prevent pooling raw observations. The present study *simulates* that constraint on the public SWAN-SF benchmark; it does not demonstrate that any specific agency’s data cannot be pooled. We present, to the best of our knowledge and per a documented search protocol (Section 2), the first federated treatment of solar flare prediction: six *simulated* regional clients—hypothetical custodians labelled by region, not participating institutions—hold disjoint, audited Dirichlet-skewed shards of the benchmark. Each client trains a local neural model on 144 statistical features derived from 24 photospheric magnetic-field parameters and transmits only weight updates. All results are produced under a frozen evaluation protocol: an immutable train/validation/test contract, calibration and threshold selection on validation only, a runtime leakage audit, and single-pass test evaluation. A data-provenance audit against the raw benchmark (Harvard Dataverse; every cleaned window aligned to its raw instance by normalization-invariant matching, 98.7–100% verification with 100% label agreement) then found that the cleaned export’s same-partition train/test pairing shares instances: the test export of each partition contains *all* raw instances while the training export is a rebalanced subset of the same instances, so 100% of the flaring test windows (6,234) and 99.998% of verified training windows were also test windows, and 85.7–90% of training positives are TimeGAN-synthetic. The study therefore re-evaluates every headline claim on a leakage-free fold that follows the benchmark’s intended temporally-preceding protocol (training partitions 1–4, single-pass test on partition 5, natural 1.31% prevalence, identical frozen protocol). On that fold, discrimination generalises for all arms (ROC-AUC 0.906–0.978), the in-partition federated-versus-centralised gap disappears (FedProx 0.976 / PR 0.308 vs centralised MLP of identical architecture 0.973 / 0.382), and FedProx retains a ROC stabilisation over plain FedAvg (0.976 vs 0.906) whose val-curve dynamics are checkpoint-independent, although FedAvg’s PR-AUC is higher (0.370 vs 0.308). Finally, retraining the identical frozen protocol on the RAW, unbalanced benchmark (natural 2.05% training prevalence, no synthetic positives, the release’s FPCKNN imputation and LSBZM normalisation reproduced in-pipeline with train-only parameters, verified against the released export) separates the arms decisively: the centralised baselines are substrate-robust (LR unchanged at 0.978; centralised MLP PR 0.445) while every federated *MLP* arm degrades sharply (FedProx 0.930/0.149; FedAvg detects zero of 65 events; SCAFFOLD fails at 0.768/0.057). The collapse is encoder-conditional: under the same protocol the sequential SolarLSTM (owner GPU) does not collapse—FedAvg-LSTM 0.958/0.305 against its MLP counterpart’s 0.875/0.056, FedProx-LSTM 0.970/0.404, above its pooled comparator though below logistic regression, and at event level FedAvg-LSTM detects 35/65 events where its MLP counterpart detects zero while FedProx-LSTM’s 49/65 comes at a centralised-level false-alarm rate and SCAFFOLD-LSTM posts the substrate’s highest detection (55/65) at 0.4 false-alarm windows per day—so the balanced-substrate standing is a property of the flattened-feature MLP, not of federation per se. A second-seed replication reproduces every sequence arm’s ranking within ± 0.005 ROC-AUC, and per-client SMOTE at natural prevalence is a clean negative (unchanged for FedAvg, -0.116 PR-AUC for FedProx); the case for federating this problem rests on data locality, not performance. Plain FedAvg diverges under the preregistered headline configuration (validation AUC $0.99 \rightarrow 0.05$ after round 20; the in-partition multiseed, sweep, ablation and client-level studies are retained as protocol-stability evidence and explicitly not as generalisation estimates). Cross-model SHAP analysis shows both the centralised and federated models concentrate decisions on current helicity, the R-value flux-emergence proxy, and Lorentz-force terms. We document a threshold-transfer failure that persists leakage-free under the prevalence shift and report prevalence-robust recall-at-FPR discrimination statistics; an event-level evaluation on 65 physical M/X-class events (raw-benchmark event keys, 1-hour alert cooldown) shows all arms detect 98.5–100% of events with a 23.4-hour median lead time to flare peak, while XGBoost’s false-alarm burden (1.7 windows/day) is roughly $7\times$ lower than every neural arm’s—the deployability boundary on this benchmark is calibration, not detection. An independent re-execution from the public dataset artefacts (byte-verified against the SHA-256 data manifest) reproduces every in-partition headline metric exactly; in the same session a five-arm calibration comparison shows that *no* validation-fit calibrator survives the prior shift, and an untouched-holdout study confirms that the proximal-stabilised global model matches local-only performance for every simulated client while clearly aiding the two smallest. The verified property is data locality, not cryptographic privacy, and the model is an upstream flare-probability component: the flare→CME→storm→GIC chain and SCADA integration remain future work.

Keywords: federated learning; solar flare prediction; smart grid; geomagnetically induced currents; FedProx; space weather; SWAN-SF; non-IID data; class imbalance; data locality

Source code and all experimental artefacts: <https://github.com/Daveshvats/Federated-Learning-for-Distributed-Space-Weather-Monitoring>

1 Introduction

Modern society has quietly built its electrical backbone in the path of a cosmic hazard. Geomagnetically induced currents (GICs) are quasi-direct currents that flow in high-voltage transmission lines when a geomagnetic storm—itsself driven by the arrival of a coronal mass ejection (CME) associated with a major solar flare—distorts the Earth’s magnetosphere. The time-varying geomagnetic field induces geo-electric fields in the conducting crust, and these fields drive GICs through grounded transformer neutrals. Once inside a high-voltage transformer, even a modest GIC biases the alternating flux into half-cycle saturation, producing harmonics, reactive-power absorption, and localized heating that can culminate in irreversible thermal damage [1, 2]. The canonical demonstration of this failure mode remains the 13 March 1989 storm, which collapsed the Hydro-Québec grid in ninety seconds and left six million customers without power for nine hours [1, 2]. The July 2012 event, an eruptive flare/CME of Carrington-class magnitude that missed Earth’s orbital position by roughly nine days, is a stark reminder that the next exposure is a matter of when, not if [3].

The first line of defence against such events is forecasting. Grid operators who receive an early warning that an M- or X-class flare has erupted, or is likely to erupt, in a magnetically complex active region can pre-emptively derate transformers, adjust reserve margins, postpone switching operations, and prepare load-shedding plans in the window between the flare observation and the CME’s arrival one to three days later. Machine-learning flare-prediction models built on photospheric vector-magnetogram summaries—the SHARP parameter suite [4] and its descendants—have demonstrated statistically skilful classification of flaring versus non-flaring active regions [5, 6]. The predictive signal lives in quantities such as total unsigned current helicity, magnetic free energy, and flux-emergence proxies, all of which are computed routinely by solar observatories on every continent and in orbit.

There is, however, a structural obstacle to training the strongest possible model: many institutions that could pool their data choose not to. The highest-fidelity flare-forecasting observations are produced and curated by nationally mandated agencies—NASA and NOAA in the Americas, ESA in Europe, JAXA in the Asia-Pacific region, ISRO in South Asia, KASI in East Asia, and the Australian Bureau of Meteorology (BoM) in Oceania—and in several jurisdictions the raw magnetogram-derived time series are treated as nationally sensitive assets whose redistribution is restricted by data-locality policies, inter-agency agreements, or export controls. Whether these restrictions are legally binding or a matter of institutional policy, the practical effect is the same: if the custodians do not pool, each either trains on its own regional shard or forgoes machine learning altogether, and the community inherits models that each see a biased slice of the active-region population. The present study does not claim that any specific agency’s flare data cannot be pooled; it takes data locality as a plausible deployment constraint and studies its consequences under a controlled simulation of that constraint.

Federated learning (FL) [7, 8] is precisely the mechanism for this setting. In cross-silo FL, each data-holder trains a shared model architecture locally and transmits only parameter updates to an aggregation server, which averages them into a new global model. Raw measurements never leave the custodian’s infrastructure. FL has already proven transformative in medicine and digital health, where patient-privacy constraints are structurally similar to the data-locality constraints just described [9]. What has not existed, prior to the present work and per the documented search protocol of Section 2, is a systematic treatment of FL for solar flare prediction—the solar-physics community’s benchmark workflows [5, 6, 10] are all centrally pooled, and we found no application of FL to photospheric flare forecasting in the FL literature.

This paper closes that gap. We present a complete, reproducible federated framework—implemented, versioned, and released as open source—that *simulates* six regional data custodians (labelled A–F by region; no real institution participates), each holding a non-IID shard of the

SWAN-SF benchmark [10] in its cleaned and rebalanced form, and evaluates whether federation can approach the centralised upper bound without any data movement. The framework is an upstream flare-probability component: it emits calibrated M/X flare probabilities and fixed-false-positive-rate alerting points that a future geomagnetic-hazard pipeline (CME/interplanetary propagation, storm forecasting, GIC and power-system response modelling) could consume; the downstream chain is not evaluated here. The specific contributions are:

1. **A problem formulation and system architecture** that maps grid-resilience precursor alerting onto a cross-silo federated classification problem with a safety-critical, recall-weighted objective, and that operationalises the data-locality constraint as a concrete, simulable protocol (Section 3 and Section 4).
2. **A non-IID federation protocol for SWAN-SF** that combines Dirichlet label-skew partitioning across six simulated regional clients, a server-aware Fed-Focal loss [11] for extreme class imbalance, and optional distribution-aware aggregation and per-client SMOTE over-sampling (both studied as ablation arms; the SMOTE arm is inert on this benchmark variant), on top of both FedAvg [7] and FedProx [12] training loops (Section 4).
3. **A frozen, leakage-audited evaluation protocol and a data-provenance audit of the benchmark export.** The protocol comprises an immutable train/validation/test contract, calibration and threshold selection on validation only, single-pass test evaluation, a centralised-MLP architecture control, five-seed statistics, a 24-cell component-ablation grid, region-disjoint split generation, natural-prevalence operating points, untouched client holdouts, a training-budget audit, event-level alerting metrics, and a calibration-comparison harness (Section 5). The provenance audit aligns every cleaned window to the raw SWAN-SF instances (Harvard Dataverse; normalization-invariant matching) and finds that the cleaned export’s same-partition train/test pairing shares instances—including 100% of the flaring test windows and 85.7–90% TimeGAN-synthetic training positives—so the study re-evaluates all headline claims on a leakage-free temporally-preceding fold (Section 5.7).
4. **A leakage-free federated evaluation.** On the temporally-preceding fold (training partitions 1–4, single-pass test on partition 5, natural 1.31% prevalence), all arms generalise (ROC-AUC 0.906–0.978); the in-partition federated-versus-centralised gap disappears (FedProx 0.976/0.308 vs architecture-matched centralised MLP 0.973/0.382); FedProx retains checkpoint-independent ROC stabilisation over plain FedAvg (0.976 vs 0.906); the threshold-transfer and calibration failures persist; and an event-level evaluation on 65 physical M/X-class events shows detection is not the deployability boundary—XGBoost’s false-alarm burden is roughly $7\times$ lower than every neural arm’s (Section 6.1).
5. **An in-partition data-locality study, disclosed as such.** On the same-partition protocol (331,185 test windows, 1.88% prevalence) whose instance overlap the audit quantified, FedProx attains ROC-AUC 0.954 (97.7% of the XGBoost reference’s 0.977) against the architecture-matched centralised MLP’s 0.888, a comparison whose optimisation budgets are documented and budget-matched, while plain FedAvg diverges during training (validation AUC $0.99 \rightarrow 0.05$); these numbers are retained as protocol-stability evidence, not as generalisation estimates (Section 6).
6. **A cross-model interpretability audit.** SHAP analysis of both the centralised XGBoost reference and the federated global model, with physical-parameter-group aggregation, shows both concentrate decisions on current helicity, the Schrijver flux-emergence proxy [13], and Lorentz-force components—consistent with established flare physics (Section 6).

We are explicit about what is *not* claimed: per-client SMOTE contributes nothing on this benchmark variant (zero synthetic samples; reported as a null result, not a contribution); the

verified property is data locality, not cryptographic privacy; the six clients are simulated, not real agencies; and the downstream flare→CME→storm→GIC chain is discussed as future integration, not evaluated.

In addition to these scientific contributions, the released framework embodies a documented engineering narrative: a multi-version stability iteration (v2.1–v3.1) in which Dirichlet concentration, focal-loss alpha clamping, optimiser choice for variance-reduced FL, and dataset auditing were each adjusted in response to diagnosed failure modes. We summarise this evolution in Appendix A because it constitutes, in our view, transferable know-how for practitioners attempting the first FL deployment in any physical-science domain.

The remainder of the paper is organised as follows. Section 2 reviews flare prediction and federated learning. Section 3 formalises the problem. Section 4 details the system architecture and methodology. Section 5 describes the experimental setup, Section 6 presents and analyses results, and Section 7 discusses operational integration and engineering lessons. Section 8 states limitations and future work, and Section 9 concludes.

2 Background and Related Work

2.1 Solar flare prediction from photospheric magnetograms

The physical premise of data-driven flare forecasting is that the photospheric magnetic field of an active region encodes the stress budget that a future flare will release. Schrijver [13] demonstrated that a single scalar—the R-value, the unsigned flux within high-gradient polarity-separation corridors—separates major-flaring regions from non-flaring ones with striking reliability, and this proxy remains among the strongest single flare predictors identified to date. Bobra and Couvidat [4] operationalised the machine-learning view of the problem, showing that a support-vector machine trained on 25 SHARP parameters derived from SDO/HMI vector magnetograms can forecast M-class flares with useful skill, and that feature selection matters as much as model choice. Subsequent community-wide blind tests, most notably the multi-method comparison of Leka et al. [5], established two durable findings: (i) no single method dominates across all event definitions and skill metrics, and (ii) the effective ceiling on skill is set by the quality and homogeneity of the training data rather than by classifier complexity.

The European FLARECAST project [6] pushed the operational agenda further, combining multiple forecasting methods and data sources in an open framework, and it explicitly confronted the data-heterogeneity problem that motivates the present paper. On the data side, the SWAN-SF benchmark of Angryk et al. [10] released a multivariate time-series dataset of roughly 8,000 active-region tracks with 24 physical parameters per timestep and flare-history labels, specifically designed to make flare prediction reproducible and comparable; it has since become the community’s reference benchmark, and a family of cleaned and rebalanced variants (random undersampling, Tomek-link cleaning, and TimeGAN-based synthetic augmentation [14]) has been curated to make the benchmark tractable for supervised learning. Recent deep-learning work on SWAN-SF continues to raise the centralised ceiling: contrastive pretraining and multifaceted preprocessing [15] and LSTM-style temporal architectures [16] exploit the sequential structure of the 60-minute observation windows. All of these works, without exception, assume that the full training corpus is available in one place.

2.2 Federated learning under statistical heterogeneity

Federated learning was introduced by McMahan et al. [7] in the FedAvg formulation: clients perform several epochs of local SGD on private data, and a server aggregates the resulting weights by sample-size-weighted averaging. The defining difficulty, identified almost immediately,

is statistical heterogeneity: when client data distributions differ—the non-IID setting—local optima diverge from the global optimum, and naive averaging can oscillate or diverge [17]. FedProx [12] addresses this by adding a proximal term $(\mu/2) \|\mathbf{w} - \mathbf{w}^{(t)}\|^2$ to each client’s local objective that anchors local training to the current global model, tolerating partial work and heterogeneous local computation. SCAFFOLD [18] attacks the same problem with control variates that correct client drift using variance reduction. The comprehensive survey of Kairouz et al. [8] situates these algorithms within the broader open-problem landscape, including robust aggregation, personalisation, and privacy. Cross-silo FL—few, reliable, institutionally owned clients—is the deployment regime relevant to space-weather agencies, and it contrasts with the cross-device regime in its stability requirements and long-lived participants.

2.3 Class imbalance in federated learning

Flare prediction inherits an extreme class-imbalance regime: in SWAN-SF’s natural test distribution, roughly 1.9% of windows precede M/X-class flares, while the rebalanced training partitions sit near parity. The focal loss of Lin et al. [19], originally developed for dense object detection, down-weights easy examples with a $(1 - p_t)^\gamma$ modulating factor and concentrates gradient on hard positives; it has become the default imbalance workhorse. Sarkar et al. [11] adapted focal loss specifically to FL in the Fed-Focal formulation, in which the class-weighting parameter is adjusted using information available at the server about the global class distribution, so that clients with different local positivity rates do not implicitly reweight the global objective. In parallel, synthetic oversampling remains a standard pre-processing response: SMOTE [20] interpolates minority-class neighbours, and mixup [21] creates convex combinations of training pairs. Under federation, rebalancing must be applied locally per client, since the pooled distribution is precisely what no participant can inspect.

2.4 Positioning of this work and search protocol

Two research threads—physics-grounded flare prediction and heterogeneity-robust federated learning—have developed largely independently, and we are not aware of any prior work that joins them. Because the first-FL-treatment claim is a factual statement about the literature, we document the search protocol behind it rather than relying on “to the best of our knowledge”. The search was last repeated on 29 September 2026 across general web, arXiv, NASA ADS, and publisher indexes (ScienceDirect, IEEE) reached through a web search engine, using four query strings: (i) *federated learning solar flare prediction*; (ii) *federated learning space weather forecasting*; (iii) *federated learning SWAN-SF benchmark magnetogram*; (iv) *cross-silo federated learning geophysics heliophysics distributed observatories*. Records were included if they apply federated or distributed multi-client learning without raw data pooling to flare prediction or space-weather forecasting; records were excluded if they concern centralised machine learning for flares (the case for all entries of query i), federated forecasting of *solar power or irradiance*, federated *terrestrial* weather forecasting, or federated computing on spacecraft without a scientific forecasting task (the categories returned by query ii). Screening the 30 records returned by the four queries yielded *zero* in-scope records; the raw result listings are committed with the repository ([docs/lit_search/](#)). We note two limitations of this protocol: web-engine coverage of grey literature is incomplete, and the screening was single-reviewer, so a PRISMA-style dual-screened log remains queued repository work. Within those limits, the present work is the first federated treatment of solar flare prediction, the first FL framework evaluated on the SWAN-SF benchmark, and the first to quantify the cost of the data-locality constraint—the gap between federated and centralised skill—specifically for a safety-critical smart-grid application. In doing so it also contributes a rarely documented artifact: an honest engineering log of the

failure modes encountered when FL meets a severely imbalanced physical-science benchmark (Appendix A).

3 Problem Formulation

3.1 Federated risk minimisation

Let K denote the number of participating agencies (clients), each holding a private dataset $\mathcal{D}_k = \{(x_i, y_i)\}_{i=1}^{n_k}$ of active-region observation windows $x_i \in \mathbb{R}^d$ and binary labels $y_i \in \{0, 1\}$ indicating whether an M- or X-class flare occurs within the forecasting horizon. Let $\mathcal{D} = \bigcup_{k=1}^K \mathcal{D}_k$ with $n = \sum_k n_k$. The goal of standard centralised learning is to solve

$$\min_{\mathbf{w} \in \mathbb{R}^p} F(\mathbf{w}) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\mathbf{w}}(x_i), y_i), \quad (1)$$

where $f_{\mathbf{w}}$ is the prediction model, ℓ a per-sample loss, and p the number of parameters. In the federated setting this objective is decomposed as a weighted sum of local empirical risks

$$\min_{\mathbf{w}} F(\mathbf{w}) = \sum_{k=1}^K \rho_k F_k(\mathbf{w}), \quad F_k(\mathbf{w}) = \frac{1}{n_k} \sum_{(x_i, y_i) \in \mathcal{D}_k} \ell(f_{\mathbf{w}}(x_i), y_i), \quad (2)$$

with aggregation weights $\rho_k \geq 0$, $\sum_k \rho_k = 1$. The vanilla FedAvg choice is $\rho_k = n_k/n$; Section 4 describes the distribution-aware weights used in this work.

3.2 The data-locality constraint

The operational constraint that distinguishes cross-silo FL from distributed optimisation is not an algorithmic preference but an institutional invariant: for every round t and client k , the server may observe only the weight increment $\Delta \mathbf{w}_k^{(t)} = \mathbf{w}_k^{(t+1)} - \mathbf{w}_k^{(t)}$ (or equivalently the locally trained weights), never x_i or y_i for any sample in $\mathcal{D}_k$. Formally, the communication transcript

$$\mathcal{T} = \{\Delta \mathbf{w}_k^{(t)}\}_{k=1, \dots, K; t=1, \dots, T} \quad (3)$$

is the only quantity that crosses the client boundary. This encodes the data-locality position that several public data custodians adopt for raw magnetogram time series, and it is the property that the released framework enforces by construction: the client-side pipeline never serialises raw shards. We stress the scope of this property: it is *data locality*, not cryptographic privacy. The transcript $\mathcal{T}$ itself can leak information about training data (Section 4.9); defending against an adversary who observes it requires secure aggregation or differential privacy, which are implemented but not evaluated for accuracy cost here.

3.3 Statistical heterogeneity

Because different custodians monitor different populations of active regions—different longitudes, hemispheres, and instrument footprints—the local class priors $p_k = \Pr(y = 1 \mid k)$ differ from the pooled prior $\bar{p}$. We simulate this label skew by partitioning the pooled training data among K clients with a symmetric Dirichlet distribution of concentration α (Section 4), so that the realised local priors span a wide range around $\bar{p}$. The consequence for optimisation is well documented [12, 17]: local objectives F_k become misaligned, and plain averaging of locally optimised weights introduces drift that grows with the number of local epochs. FedProx counteracts this drift by locally solving

$$\min_{\mathbf{w}} F_k(\mathbf{w}) + \frac{\mu}{2} \|\mathbf{w} - \mathbf{w}^{(t)}\|^2, \quad (4)$$

which keeps client updates in a trust region around the global model. The strength of this anchor, μ , is a central hyperparameter of this study.

3.4 A safety-critical, recall-weighted objective

Grid protection inverts the usual classification cost structure. A false-positive flare alert costs a operator a conservative derating or an unnecessary operational posture; a false negative can cost a fleet of high-voltage transformers. Following operational forecasting practice, we therefore evaluate with the F_β measure

$$F_\beta = \frac{(1 + \beta^2) P R}{\beta^2 P + R}, \quad \beta = 2, \quad (5)$$

where P and R are precision and recall; $\beta = 2$ weights recall twice as heavily as precision. Because the deployed decision rule is a threshold τ on the predicted flare probability, the framework treats τ as a free, per-model parameter selected on held-out validation evidence to maximise Eq. (5), rather than fixing $\tau = 0.5$ by convention. Section 6 shows this choice is not cosmetic: at the conventional threshold, every federated model trained on rebalanced local data is operationally unusable, and threshold optimisation is what converts a well-ranked model into a usable detector.

4 System Architecture and Methodology

4.1 Architecture overview

Figure 1 shows the complete system. Six simulated regional clients—labelled neutrally A–F by region (Americas, Europe, Asia-Pacific, South Asia, East Asia, Oceania); no real institution participates in this study—each hold a private, non-IID shard of the benchmark. In every communication round, each selected client trains a local copy of the shared model on its shard (with an imbalance-robust loss, described below), and returns only its weight increment to the aggregation server. The server forms the next global model by size-weighted averaging, broadcasts it to all clients, and the cycle repeats for $T = 50$ rounds. Downstream of federation, the global model emits flare probabilities and, at validation-frozen operating points, M/X flare-precursor alerts: an *upstream* component that a future grid-resilience pipeline—CME propagation and storm forecasting, geomagnetic-hazard estimation, GIC and power-system response modelling, and SCADA integration—could consume. That downstream chain is outside the scope of this paper and is not evaluated here (Section 8). The division of labour is deliberate: everything that touches raw magnetogram data (loading, local training) executes inside the client boundary; everything the server touches is a weight tensor.

4.2 Data: the SWAN-SF benchmark and its cleaned variant

All experiments use the SWAN-SF benchmark [10] (doi:10.7910/DVN/EBCFKM), a multivariate time-series dataset of active-region observation windows drawn from SDO/HMI vector magnetograms, with 24 physically motivated parameters per timestep sampled over 60-minute sliding windows and labels indicating the occurrence of $\geq M$ -class flares within the prediction horizon. The 24 parameters summarise magnetic-field complexity and include, among others: total unsigned current helicity (TOTUSJH), total photospheric magnetic free energy (TOTPOT), total unsigned vertical current (TOTUSJZ), total unsigned flux (USFLUX), the sum of absolute net-current-per-polarity (SAVNCPP), absolute net current helicity (ABSNJZH), mean shear angle and the fraction of high-shear area (MEANSHR, SHRGT45), the Schrijver flux-emergence proxy

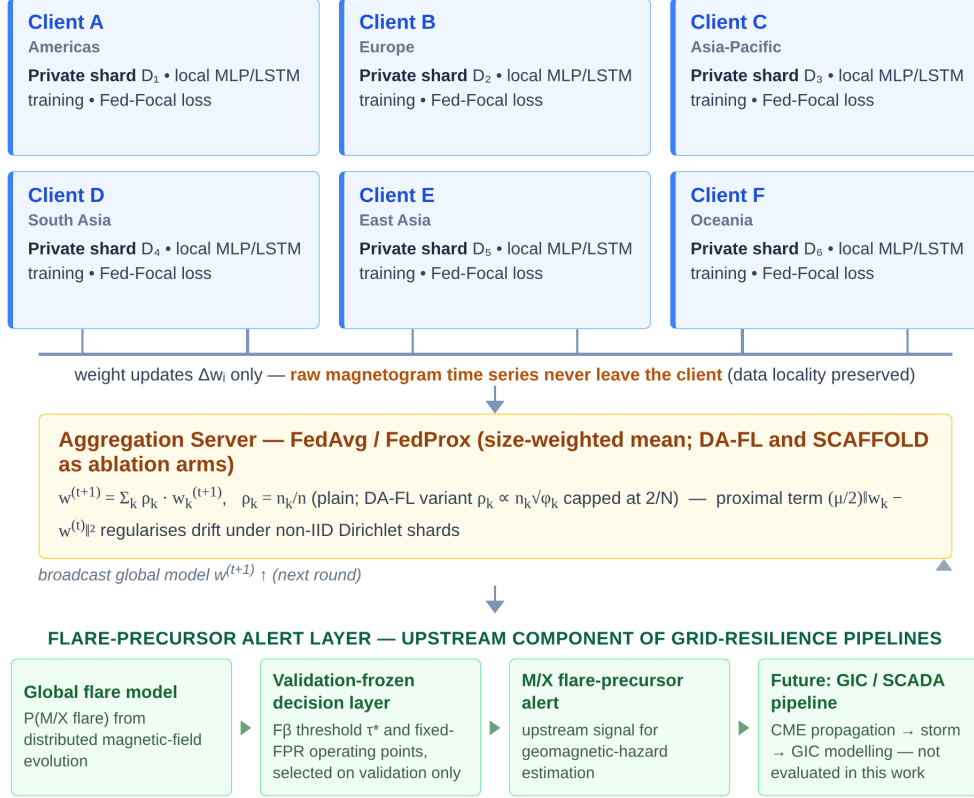


Figure 1: Federated space-weather monitoring architecture for grid-resilience pipelines. Six simulated regional clients (neutral labels A–F; no real institution participates) hold private non-IID shards of the benchmark; only weight updates Δw_k cross the client boundary. The server applies size-weighted FedAvg/FedProx aggregation by default (the distribution-aware weighting $\rho_k \propto n_k \sqrt{\phi_k}$, capped at $2/K$, is studied as an ablation arm). The global model emits flare-probability precursor alerts at validation-frozen operating points; the downstream GIC/SCADA pipeline is a future integration interface and is not evaluated.

(R-value), and the three components of the total Lorentz force (TOTFX, TOTFY, TOTFZ) [4, 13].

We ingest the benchmark through its community-maintained cleaned variant¹, combining all five data partitions. The training partitions are class-balanced by this preprocessing (pooled positive rate $\bar{p} \approx 0.489$), while the untouched test partitions retain the natural, severely imbalanced prevalence of approximately 1.9% flare windows across roughly 3.3×10^5 held-out samples. This train–test contrast is not an artefact we introduce; it is inherited from the benchmark’s design philosophy, and it has a consequence that shapes the entire evaluation: models trained at $\sim 49\%$ prior must detect events at a $\sim 2\%$ prior, so absolute probability calibration degrades and decision thresholds must be re-optimised post hoc (Section 4.8).

The raw-benchmark experiments of Sections 6.1 and 6.2 additionally ingest the unprocessed SWAN-SF partitions directly. The raw substrate cannot be trained on as-is: 8.87% of its values are missing, the missingness is structured (the Lorentz-force component TOTFZ is unobserved in 98.7% of training windows), and the 24 parameters span eleven orders of magnitude. We therefore reproduce the cleaned release’s two preprocessing stages in-pipeline from their published description (ApJS, 10.3847/1538-4365/ad7c4a) rather than inheriting their

¹Cleaned SWAN-SF Dataset, <https://github.com/samresume/Cleaned-SWANSF-Dataset>, which applies per-partition preprocessing: random undersampling with Tomek links and TimeGAN-based synthetic augmentation for the training partitions [14], LSBZM normalisation, and FPC-KNN imputation.

output: a fast Pearson-correlation-based k-NN imputation (within-window linear interpolation for partial gaps; inverse-distance-weighted transfer of the $k=5$ nearest training windows’ series, searched in the space of the five most-correlated observable features, for fully-missing series), and the LSBZM normalisation chain (robust quantile clip, positivity shift, a log/sqrt/Box–Cox stabiliser dispatched per feature by fitted λ , z-score, min–max), with every parameter fitted on the training pool only and a per-feature power-of-two pre-scaling for numerical safety. The reproduction is verified against the released export on the audit-matched windows (observed-position rank agreement in Section 6.2); where the two pipelines differ substantively (the release re-imputes the R-value zero mass as if missing; we preserve zeros as physical “no flux emergence” values), we preserve the raw semantics. The construction, the runner, and the verification are committed as `experiments/raw_substrate.py`, `experiments/run_raw_substrate.py`, and `experiments/run_event_level_raw.py`.

4.3 Client-side feature representation

The framework supports two representations of the same 60-timestep observation windows. In the MLP path used for the reported run, each three-dimensional window $X \in \mathbb{R}^{60 \times 24}$ is reduced to a 144-dimensional statistical vector

$$\tilde{x} = [\mu, \sigma, \max, \min, \delta, s] \in \mathbb{R}^{144}, \quad (6)$$

where each block is applied per base feature: μ and σ are the temporal mean and standard deviation, $\max$ and $\min$ the window extrema, $\delta = X_T - X_1$ the net trend, and s the least-squares slope against normalised time. This six-statistic extraction preserves substantially more temporal information than a temporal mean (peaks, volatility, and rate of change are precisely the flare-relevant quantities) while remaining compatible with weight-averaged federation and with the centralised baselines. In the LSTM path, the raw $\mathbb{R}^{60 \times 24}$ tensor is consumed directly by a recurrent model; this path is fully implemented in the released code but the committed artefacts analysed here correspond to the MLP execution, and we flag the LSTM evaluation as future work (Section 8).

4.4 Non-IID partitioning into regional observatories

To emulate geographic heterogeneity, the pooled training set is partitioned by label-aware Dirichlet sampling: for each class c , a proportion vector is drawn from $\text{Dirichlet}(\alpha, \dots, \alpha) \in \mathbb{R}^K$ and each client receives the corresponding fraction of that class. The concentration α controls heterogeneity strength—small α produces near-exclusive class ownership; large α approaches IID. Figure 2 shows the deterministic realisation generated by the released configuration ($\alpha = 1.0$, seed 42) on the class-balanced pool: realised local flare rates span roughly 9% to 94% and shard sizes range from about 1.5% to 38% of the pool. This is a deliberately hostile federation: some clients are flare-rich and tiny, others flare-poor and large, mirroring the real asymmetry between agencies whose instruments and coverage dominate the corpus and those monitoring smaller regional populations. The concentration $\alpha = 1.0$ itself is the outcome of a documented tuning process (Appendix A): early experiments at $\alpha \leq 0.5$ produced pathological partitions (0.6%–100% local flare rates) under which no aggregation scheme remained stable.

4.5 Client models

The primary federated architecture is a four-layer multilayer perceptron, $\text{SolarMLP} : \mathbb{R}^{144} \rightarrow \mathbb{R}$, with hidden widths $144 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$, batch normalisation, ReLU activations, and

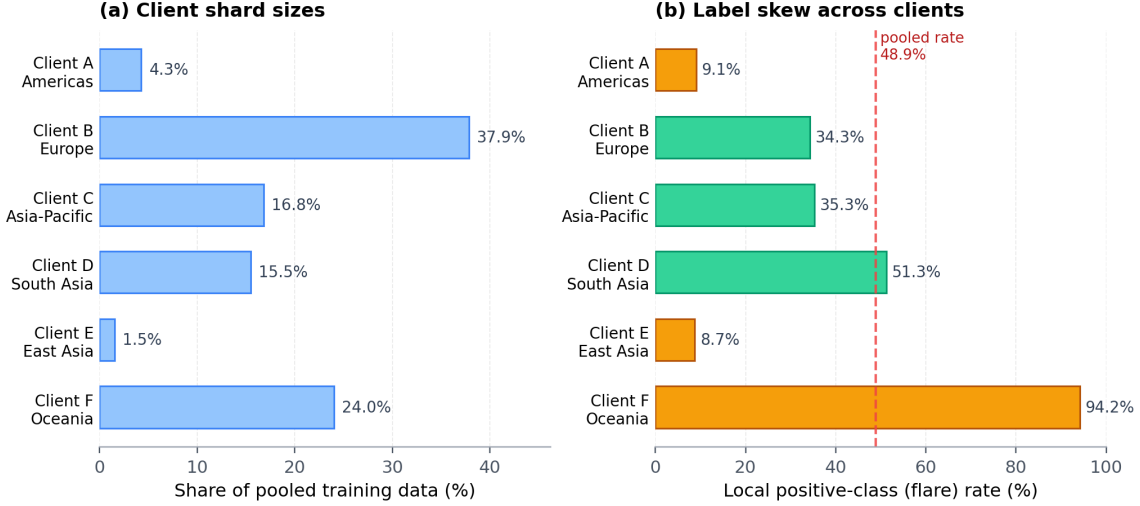


Figure 2: Non-IID partitioning of the pooled, class-balanced training data into six regional clients by symmetric Dirichlet sampling (concentration $\alpha = 1.0$, RNG seed 42, Section 4). (a) Realised relative shard sizes. (b) Realised local flare rates against the pooled rate of 48.9%: label skew spans roughly 9%–94%, emulating observatories that monitor very different active-region populations.

dropout 0.3 after each hidden layer; the single output is a raw logit consumed by the sigmoid-based losses below. MLPs are natural federation citizens because every parameter is a numerically meaningful averageable tensor. The released framework also implements SolarLSTM, a (optionally bidirectional) two-layer LSTM with hidden size 128, Xavier/orthogonal initialisation, and a forget-gate bias of one for long-horizon memory, followed by a two-layer classification head [16, 22]. Local training uses AdamW [23] with cosine-annealing warm restarts and gradient-norm clipping at 5.0. We deliberately keep XGBoost out of the federation loop: gradient-boosted tree ensembles store discrete structure that cannot be meaningfully averaged, so the centralised XGBoost serves as the pooled upper bound rather than a federated participant.

4.6 Federation algorithms and aggregation

Three cross-silo algorithms are implemented. **FedAvg** [7]: each round, every client initialises from the broadcast global model, performs $E = 10$ local epochs, and the server averages the returned weights. **FedProx** [12]: identical to FedAvg except each client’s local objective is regularised by the proximal term of Eq. (4) with $\mu = 0.01$, penalising drift from the global model—the mechanism we hypothesise is decisive under the label skew of Figure 2. **SCAFOLD** [18]: variance reduction via client and server control variates; it is implemented with the SGD-based update required for exactness of the control-variate correction, but is disabled in the released configuration pending stability work, and we exclude it from the reported comparison (Section A).

Aggregation is *plain size-weighted averaging* in the frozen headline configuration ($\rho_k = n_k/n$, the vanilla FedAvg choice); distribution-aware weighting (DA-FL), in which the server weights client k ’s update by

$$\rho_k \propto n_k \sqrt{\phi_k}, \quad \phi_k = \frac{p_k}{\bar{p}}, \quad \rho_k \leq \frac{2}{K}, \quad (7)$$

with p_k the client’s local positive rate and $\bar{p}$ the pooled rate known to the server, is implemented as an *ablation arm* (Section 6.11). The square-root damping and the $2/K$ cap prevent a flare-rich client from dominating the global model while still up-weighting the clients most likely to contain decision-relevant minority samples. Class-rate metadata is deliberately much coarser than raw data and is treated as shareable under the data-locality constraint.

4.7 Fed-Focal loss and local rebalancing

Each client trains with a server-aware focal loss in the Fed-Focal family [11],

$$\ell_{\text{fedfocal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t), \quad \gamma = 2.0, \quad (8)$$

where p_t is the model’s probability of the correct class and α_t interpolates between α and $1 - \alpha$ by label. The positive-class weight α is computed per client as the base value 0.25 [19] scaled by a square-root-damped imbalance factor $\min(\sqrt{\bar{p}/p_k}, 1.5)$ and a mild round-progress factor, then clamped to (0.05, 0.25). This clamping is a hard-won design decision: unclamped adaptive weighting (earlier versions allowed α up to 0.95) systematically inflated positive-class weight on flare-poor clients, producing degenerate always-flare classifiers with recall ≈ 1.0 and precision ≈ 0.03 (Appendix A). NaN-safe clamping of p_t and of the loss itself guards the pipeline against logit explosions in early rounds. The framework additionally supports per-client SMOTE oversampling [20] and mixup [21]; both are *disabled* in the frozen configuration because the cleaned benchmark’s training partitions arrive pre-balanced (48.9% positive), which renders SMOTE inert—the configured oversampling ratio produces zero synthetic samples. The balancing question is therefore inapplicable to this benchmark variant, and we report it as a null result rather than a contribution (Section 6.11); SMOTE is a live arm for the raw-substrate programme, where natural prevalence finally gives it work to do (Section 8).

4.8 Threshold protocol for safety-critical recall

Because federated training operates on rebalanced local data while deployment operates at the natural 1.9% prior, the emitted probabilities are deliberately miscalibrated, and a fixed 0.5 threshold is meaningless. The frozen protocol therefore performs all selection on validation data, in two stages. First, the decision threshold is selected by sweeping τ over $[0.05, 0.95]$ in steps of 0.005 and choosing, per model, the τ^* that maximises F_β with $\beta = 2$ (Eq. (5)) on the validation set; τ^* is then frozen and applied to the test set exactly once. All reported precision, recall, F1, and accuracy figures are taken at each model’s frozen threshold. Second—because balanced-prevalence validation turns out not to support threshold transfer at all (Section 6.9)—the framework additionally freezes *fixed-FPR operating points*: for each target false-positive budget $f \in \{0.5\%, 1\%, 2\%, 5\%\}$, the threshold is set to the $(1 - f)$ -quantile of *validation negative scores*, a quantity that uses no test information whatsoever; the realised test behaviour at each frozen threshold (achieved FPR, recall, precision, alert rate) is reported in the same single test pass. The negative-quantile rule is numerically validated in the released test suite, and a natural-prevalence validation split, the ideal substrate for this protocol, is part of the queued protocol re-run (Section 5). ROC-AUC and PR-AUC, being threshold-free, are reported independently.

4.9 Threat model and security posture

The verified property of this framework is data locality (raw data never leaves the client), not end-to-end security. To make that scope auditable, Table 1 states the threat model explicitly: which adversaries are considered, which defences the released code provides, and which are documented but unevaluated.

For the benchmark study reported here—six simulated clients, public data, no real adversaries—none of these defences is enabled, so the reported accuracy numbers are those of the undefended pipeline. Any cross-agency deployment would need to re-measure accuracy under secure aggregation and differential privacy before the numbers in Section 6 could be quoted as expected operational performance (Section 8).

Table 1: Threat model of the released framework. “Mitigated in code” means implemented and covered by the integrity test suite; “documented” means design guidance exists but no quantitative evaluation has been performed on this benchmark.

Adversary / attack	Defence	Status
Honest-but-curious aggregation server (observes all weight updates)	Secure-aggregation module (Bonawitz-style masks; round-trip verified by tests)	Implemented as verified module; not yet wired into the training loop; accuracy overhead not measured
Gradient inversion / reconstruction from the transcript $\mathcal{T}$ [24]	Secure aggregation (masks updates); differential-privacy hooks	Documented; not evaluated
Membership inference against a specific client’s training data	Differential-privacy hooks (noise on client updates)	Documented; not evaluated
Malicious client: model poisoning / backdoor insertion	None at present; Byzantine-robust aggregation is a design note	Not addressed
Client impersonation / spoofed updates	Transport security (TLS) and client authentication are deployment infrastructure outside the learning loop	Documented (deployment guidance)
Network interception of updates	TLS in transit	Documented (deployment guidance)

4.10 Interpretability layer

To audit what the models learn, SHAP values [25] are computed for the centralised XGBoost reference over a 500-sample test subset with the TreeExplainer and for the federated model with DeepSHAP (100-sample background), yielding mean absolute SHAP attributions per feature. The statistical representation of Eq. (6) carries a 144-entry name contract—each feature is named `<statistic>_<parameter>` (e.g. `mean_TOTUSJH`, `max_R_VALUE`), with the statistic blocks ordered μ (0–23), σ (24–47), $\max$ (48–71), $\min$ (72–95), δ (96–119), s (120–143) over the 24 base parameters—and the improved pipeline preserves these names end-to-end, so the SHAP layer reports physically meaningful attributions directly (the v2.x pipeline lost the mapping and required an analytical decoding pass; Appendix A). Section 6 presents the raw ranking, its statistic-level structure, and the physical grouping used for cross-model comparison.

5 Experimental Setup

5.1 Frozen evaluation protocol

All experiments in this revision run under an immutable evaluation contract, enforced in code before any training is permitted:

- **Splits.** The benchmark’s five training partitions (97,764 windows, balanced, 48.87% positive) form the training pool; the five test partitions (331,185 windows, natural prevalence, 1.88% positive) are the held-out test set and are loaded exactly once at the very end of the pipeline. A stratified validation set (16%, $n=15,643$) is carved from the *training* pool with the root seed (42) and is the only set used for any model selection.
- **Selection isolation.** Hyperparameter monitoring, the best-checkpoint rule, calibration fitting, the F_β threshold search, and the fixed-FPR operating-point thresholds all operate on

validation. The test set participates in no selection of any kind (the v2.x pipeline selected thresholds directly on test; this revision removes that violation).

- **Leakage audit.** A runtime audit verifies sample-disjointness of train/validation/test, disjointness and coverage of the client shards, and flags the known limitation that active-region identities are not available in the cleaned export, so region-level separation across the train/validation boundary cannot be guaranteed (Section 8). The pipeline refuses to train if the audit fails.
- **Single-pass evaluation.** Each model’s frozen pipeline (calibrator + threshold) is applied to the test set once; all reported test metrics are recomputed from those predictions (the v2.x code reported a stale accuracy for federated models).
- **Protocol extensions released with this revision.** The following components are implemented, unit-tested (78-check integrity suite), and smoke-validated. Five of them were additionally *executed* end-to-end in the independent re-execution of Section 5.6 and their results are reported in Section 6: (ii) the validation-frozen fixed-FPR operating-point protocol of Section 4.8 (realised test behaviour in Section 6.9); (iii) the untouched-holdout client evaluation mode, in which every client reserves a fraction its models never train on, so local versus global comparisons are clean generalisation estimates (Section 6.12); (iv) the budget-matched centralised retraining (epoch cap raised to 500 in two early-stopping variants; Section 5.5); (vi) the calibration-comparison harness (raw, prior-shift, Platt, isotonic, and temperature arms, validation-fit and single-pass test evaluation; Section 6.9); and the $\alpha=5.0$ promotion run at the full 50-round protocol (negative result; retained $\alpha=1.0$; Section 5.3). The remaining extensions are released and awaiting full-data execution, blocked on data that the cleaned export does not carry: (i) a region-disjoint train/validation split generator that partitions by active-region identifier when available (the cleaned export lacks HARP/AR identifiers, so the committed split remains the stratified carve-out); and (v) an event-level evaluation module (detection rate, missed events, duplicate alerts, warning lead time, alert rate per day; requires event keys absent from the cleaned export).

5.2 Baselines

Four centralised references span the difficulty range of the task. **Climatology** predicts the training prevalence for every window —the skill floor. **Logistic regression** uses balanced class weights and 2,000 maximum iterations. **Centralised MLP** is the architecture-controlled reference: the identical SolarMLP (29,377 parameters), trained on the pooled features with identical optimisation, whose comparison against the federated MLP isolates the effect of federation from the effect of architecture (the v2.x manuscript lacked this baseline and compared federated MLPs only against XGBoost). **XGBoost** [26] (300 estimators, depth 6, learning rate 0.05, subsample 0.8) is the strongest reference baseline; its `scale_pos_weight` (1.05) is now derived from the *training* prevalence rather than hand-set with knowledge of the test imbalance. Every baseline receives the same validation-selected calibration and threshold treatment as the federated models, so the decision layer is uniform across all comparisons.

5.3 Federated configuration and preregistration

Table 2 summarises the frozen configuration. The headline runs use plain size-weighted FedAvg aggregation (the DA-FL weighting of Eq. 7 is studied as an ablation arm in Section 6.11); per-client SMOTE is disabled because the balanced benchmark renders it inert. The proximal coefficient is $\mu = 0.01$. Two facts govern the heterogeneity parameter: the headline run and *all* statistical studies (multiseed, ablations, client-level evaluation) use the preregistered Dirichlet $\alpha = 1.0$; separately, the μ/α sensitivity sweep ($\mu \in \{0.001, 0.01, 0.1\}$, $\alpha \in \{0.5, 1.0, 5.0\}$, 15

Table 2: Frozen configuration of the improved framework (`improvements` branch, v3.1).

Component	Setting	Notes
Federation	$K = 6$ clients (all selected each round), $T = 50$ rounds	headline
Local optimisation	10 local epochs, AdamW, lr 5×10^{-4} , wd 10^{-4} , cosine warm restarts, batch 512, grad-clip 5.0	unchanged
FedProx	$\mu = 0.01$	sweep-validated
Aggregation	plain size-weighted mean (DA-FL: ablation)	v3.0 change
Loss	Fed-Focal, $\gamma = 2.0$, α clamped (0.05, 0.25); global prevalence computed, not hardcoded	B13 fix
Local rebalancing	SMOTE off (inert: pre-balanced data)	ablation
Partitioning	Dirichlet $\alpha = 1.0$, disjoint by construction, seed 42	B2 fix
MLP	144 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 1, BatchNorm, ReLU, dropout 0.3, 29,377 params	
Checkpointing	restore best validation F1 round	B3 fix
Calibration	prior-shift (validation-fit; Platt/isotonic/temperature available)	
Decision layer	F_2 threshold grid $\tau \in [0.05, 0.95]$ on <i>validation</i>	B1 fix
Multi-seed	seeds 42–46, mean/std/Ci95, Wilcoxon paired	new
Ablations	$2 \times 3 \times 2 \times 2$ grid, 20 rounds	new

rounds) selected $\alpha = 5.0$, $\mu = 0.01$ as the validation-optimal configuration (validation PR-AUC 0.993; its single test pass gave PR-AUC 0.297, consistent with the headline FedProx result). The two values answer different questions— $\alpha = 1.0$ is the adversarially heterogeneous *study regime*, $\alpha = 5.0$ is the validation-selected *tuning optimum*—and we do not silently substitute one for the other. The promotion experiment was executed in this revision at the full 50-round frozen protocol (`outputs/alpha5_promotion.json`): FedProx at $\alpha = 5.0$ reaches test ROC-AUC 0.955 / PR-AUC 0.296, statistically indistinguishable from the $\alpha = 1.0$ headline (0.954/0.307), so the sweep optimum does not transfer to the study regime and $\alpha = 1.0$ is retained as the preregistered headline configuration. The same run isolates *why* plain FedAvg diverges at $\alpha = 1.0$: at $\alpha = 5.0$ the identical FedAvg arm completes all 50 rounds without collapsing (test ROC-AUC 0.915, PR-AUC 0.201), i.e. the divergence is a heterogeneity-severity effect rather than an intrinsic failure of plain averaging on this task. Both values, and the provenance of each, are recorded in the run manifest.

5.4 Metrics

We report precision, recall, F1, F_2 , ROC-AUC, PR-AUC, Brier score, expected calibration error (ECE), and recall at fixed false-positive budgets (0.5%, 1%, 2%, 5%). Given the $26\times$ validation-to-test prevalence shift and the threshold-transfer failure documented in Section 6.9, the threshold-independent measures (ROC-AUC, PR-AUC, recall-at-FPR) carry the discrimination claims, while the threshold-conditional measures document the decision layer honestly. Confusion matrices are row-normalised. For the multi-seed study we report mean $\pm$ standard deviation with Student- t 95% confidence intervals and one-sided Wilcoxon signed-rank paired tests.

5.5 Optimisation budgets: centralized versus federated

The centralised-MLP versus federated comparison is only interpretable if the optimisation budgets of the two arms are stated explicitly, so the pipeline now emits a machine-readable budget audit (`training_budget` in `results.json`). Table 3 summarises it. The shipped budgets are *not* matched: the centralised reference executes at most 30 full passes over the pooled training set (with early stopping on validation loss), whereas each federated client’s shard receives

Table 3: Training-budget audit of the two architecture-matched arms (review finding: the comparison is confounded by optimisation budget; the table makes the confound auditable).

	Centralised MLP	Federated (FedProx)
Epochs / rounds	≤ 30 epochs	50 rounds $\times$ 10 local epochs
Batch size	256	512
Learning rate (peak)	5×10^{-4}	5×10^{-4}
LR schedule	cosine warm restarts	cosine warm restarts
Effective passes over own data	≤ 30 (pooled set)	500 (per client shard)
Early stopping	patience 5 (validation BCE)	none (fixed rounds)
Checkpoint rule	best validation BCE	best validation F1 round
Budgets matched	Shipped: no; matched 500-epoch re-run executed—see text	

$50 \times 10 = 500$ local passes over the 50 rounds. The budget-matched re-run was executed in this revision (`outputs/budget_matched.json`): the centralised MLP was retrained on the same pooled data with the epoch cap raised to 500, in two variants—(a) early stopping retained, which fires at epoch 13 and reproduces the headline numbers exactly, and (b) the FL-style fixed-budget protocol (no early stop, best-validation-loss checkpoint retained), which reaches test ROC-AUC 0.917 / PR-AUC 0.219. Two conclusions follow. First, the confound ran *against* the federated arm: the shipped centralised reference was *undertrained* (early stopping fires long before the budget binds), and the pooled model improves when granted the federated arm’s budget. Second, the improvement is not sufficient—at full budget parity the centralised MLP (0.917/0.219) remains below FedProx (0.954/0.307) on the threshold-free metrics, so the headline gap is a genuine federation effect on this benchmark rather than a budget artifact (its F_1 of 0.135 versus FedProx’s 0.042 reflects threshold-transfer behaviour under prior shift, not ranking skill). The architecture comparison (identical 29,377-parameter MLP) remains matched by construction. The raw-substrate LSTM arms (Section 6.3) inherit the same asymmetry: the pooled SolarLSTM comparator is capped at 30 epochs with patience-5 early stopping (which did not fire: epoch 30, validation BCE 0.039), while the federated arms receive 500 shard passes. The FedProx-LSTM-over-central-LSTM margin (+0.008 ROC-AUC) is therefore budget-caveated—the cleaned-fold parity experiment moved the central MLP by +0.029 ROC-AUC, the same order—whereas the load-bearing contrast of that section (federated LSTM versus federated MLP under identical budget and aggregation) is budget-matched by construction.

5.6 Compute and reproducibility

The complete experimental programme of this revision (headline run, 24-cell ablation grid, 9-configuration sweep, 5-seed study, interpretability, and communication measurements) was executed on a 2-core CPU-only container with under 4 GB of RAM, in aggregate approximately 2.5 hours of wall-clock time. Every experiment writes a machine-readable results artefact (`results.json`, `ablation_results.json`, `sweep_results.json`, `multiseed_results.json`, `interpretability.json`, `smote_validity.json`) plus a run manifest embedding the code version, seeds, configuration, and dataset manifest hash; all figures and tables are generated from these artefacts. The pipeline supports round-level crash recovery, and the 78-check integrity test suite (partitioning disjointness, leakage audit, calibration, evaluation protocol, threshold-freezing arithmetic, event-level metrics, FL smoke tests) passes on every commit.

Independent re-execution from the public artefacts. Following the review of v3.0.1, the entire headline pipeline was re-executed from scratch in a clean environment: the Cleaned SWAN-SF release was downloaded from the dataset repository’s official distribution, all 20 partition files were verified byte-identical to the frozen data manifest (SHA-256 match on every

file), and the frozen protocol was run end-to-end (data load and flattening, immutable splits, Dirichlet partition, pooled baselines, FedAvg and FedProx to 50 rounds, validation-fit calibration and thresholds, single-pass test evaluation, figures, and SHAP attribution) on CPU only (Python 3.12, PyTorch 2.14). Every test-set metric of every model reproduced *exactly*, to all reported significant digits: all six models’ ROC-AUC, PR-AUC, Brier, ECE, precision, recall, F1, and accuracy match the published artefacts at three decimals, and the FedAvg divergence signature reproduces round-for-round. The pipeline is therefore bit-stable under this protocol: fixed seeds, deterministic CPU kernels, disjoint index arithmetic, and single-precision training make the published numbers re-derivable by anyone from the public dataset and the released code, which is the strongest form of reproducibility claim this study can make. The same re-execution session ran, for the first time on real data, the calibration-arm comparison (Table 12), the untouched-holdout client evaluation (Section 6.12, `client_holdout_eval.json`), and the frozen-FPR operating points now reported in Section 6.9; the multi-seed, sweep, and ablation studies were not re-executed and their numbers remain the owner-side artefacts embedded in the repository.

Owner-GPU execution of the sequence arms. The raw-substrate LSTM arms of Section 6.3 were executed on owner-side GPU hardware (an NVIDIA RTX 4060 Laptop with CUDA and cuDNN), because the CPU-only protocol above would have needed an estimated 74 h—the one experiment where GPU was a practical requirement. Two batches: the seed-42 arms (2.6 h total: 0.08 h pooled comparator, 1.16 h FedAvg-LSTM, 1.33 h FedProx-LSTM) and, as a second single resumable queue, the remaining programme (8.7 h: 1.5 h SCAFFOLD-LSTM, 3.9 h the seed-43 replication of all four arms, 3.2 h the two SMOTE ablation arms). The substrate identity is device-independent and asserted at startup on every run (label equality with the frozen 2D split; 144-statistic cross-check at 4.4×10^{-4}), and each run writes the same machine-readable artefact schema as every CPU arm (`raw_lstm_eval`, `raw_lstm_scaffold`, `raw_lstm_seed43`, and `raw_lstm_smote`, all under `outputs/`). Each GPU batch was validated on receipt, before integration, by a machine artefact audit (64 checks for the second batch: protocol fingerprints, substrate identity, embedded frozen-counterpart bit-matches, event-key reconciliation, and cross-checks against the run logs). The residual device effect is kernel-level nondeterminism in cuDNN recurrence and float-reduction order, predicted within ± 0.001 – 0.005 on AUC-scale metrics and now measured by the seed-43 replication at ≤ 0.005 ROC-AUC on every arm (Section 6.4); all other artefacts in this paper were produced under the bit-stable CPU protocol above.

5.7 Data-provenance audit and the leakage-free protocol

Raw-benchmark alignment. The Cleaned SWAN-SF export carries no region or event identifiers, so the review-programme items that require them (region-disjoint splits, event-level evaluation, natural-prevalence substrates) were queued pending raw metadata. We obtained the raw benchmark from its official distribution (Harvard Dataverse, doi:10.7910/DVN/EBCFKM, 6.5 GB across the five partition archives plus the 34 MB addenda with the GOES flare event lists) and aligned every cleaned window to its raw instance. Each raw instance is a 60-timestep CSV whose filename encodes the flare class, the *HARP* region identifier, and the window start/end timestamps, and the folder (FL for M/X-flaring, NF otherwise) gives the label. The matching key is the per-feature argmax/argmin position along the 60 timesteps—48 discrete invariants that survive any per-feature monotone normalization and are robust to the export’s imputation noise. A window is accepted as aligned when at least 28 of 48 invariants agree (borderline cases require a mean per-feature Spearman rank correlation of at least 0.70). On the test partitions this verifies 98.7–100.0% of windows with 100.00% label agreement and a median best-to-second-best margin of 31–32 invariants; the alignment artefacts and per-window metadata tables are released with the code (`outputs/dataset_structure_audit.json`).

What the audit found. Three structural facts follow. First, the test export of partition p contains *all* raw instances of p (the counts match exactly on every partition). Second, the training export of p is a RUS-Tomek-TimeGAN rebalanced *subset of the same instances*: of the 56,006 verified training windows across the five partitions, 56,005 (99.998%) are also test windows, and every M/X-flaring test instance—6,234 of 6,234, 100% of the minority class—is in the training pool. Third, 85.7–90.0% of the training positive windows are TimeGAN-synthetic (they have no raw counterpart), while virtually no negative windows are synthetic. The release paper of the cleaned dataset prescribes pairing *temporally preceding* training partitions with a later test partition; the shipped pipeline of this study instead pooled the training exports of all five partitions and evaluated on the test exports of the same five, pairing each test partition with a rebalanced subset of itself. The published in-partition numbers therefore partially measure memorised data, most severely for the minority class. Because no region identifier spans two partitions (verified: 0 shared HARPs), a temporally-preceding fold is automatically instance-, region-, and time-disjoint.

Leakage-free fold. All headline-model comparisons were re-run under the benchmark’s intended protocol: training on the training exports of partitions 1–4 (77,865 windows; 65,406 train / 12,459 validation after the identical stratified carve), and a single-pass evaluation on the test export of partition 5 (75,365 windows at the natural 1.31% prevalence). Everything else is frozen: Dirichlet $\alpha = 1.0$, six clients, 50 rounds, seed 42, Fed-Focal loss, prior-shift calibration and validation-frozen thresholds. The runner is `experiments/run_partition_disjoint.py` (round-resumable); its artefacts are `outputs/partition_disjoint_eval.json` and, for the event-level analysis, `outputs/event_level_p5.json` with the per-window event keys recovered from the raw filenames and the addenda GOES peak list (65 of 65 partition-5 flare events matched, 4 with multiple candidates). Event grouping uses the filename flare tag for positive windows (one group per physical flare) and region-day groups for background windows; alert de-duplication applies a 1-hour cooldown within each group.

6 Results and Analysis

All numbers in this section are produced by the improved pipeline (improvements branch) under the frozen evaluation protocol of Section 5: a single immutable train/validation/test split, thresholds and calibration selected on validation only, and the test set touched exactly once per model. Every table and figure is generated from the machine-readable results-json artefacts of the runs; no value is transcribed by hand. The headline run uses 50 communication rounds, six clients, and seed 42; Sections 6.10 and 6.11 quantify seed sensitivity and component attribution. An independent re-execution of the headline pipeline from the public, byte-verified dataset artefacts reproduces every test metric of every model exactly (Section 5.6); the calibration, frozen-FPR, and client-holdout results below were produced in that re-execution session.

Protocol disclosure (v3.4). The data-provenance audit of Section 5.7 established that the in-partition train/test pairing used by the shipped headline protocol shares instances between the training pool and the test set—including 100% of the flaring test instances. The subsections that follow the present one therefore report *in-partition* results: they remain valid as evidence of protocol stability, seed sensitivity, component attribution, and calibration behaviour under the frozen pipeline, but they are not estimates of generalisation to unseen data. The leakage-free evaluation under the benchmark’s intended temporally-preceding protocol is reported first, and the claims of this study are tiered against it in Section 6.5.

Table 4: Leakage-free fold (train partitions 1–4, single-pass test on partition 5, natural 1.31% prevalence) against the shipped in-partition numbers. Threshold-conditional calibration metrics (Brier, ECE) use the validation-frozen prior-shift pipeline; ROC/PR are threshold-free.

Model	In-partition (shipped)		Leakage-free fold				
	ROC	PR	ROC	PR	Brier	ECE	R@2%FPR
Logistic regression	0.818	0.114	0.978	0.455	0.375	0.477	0.695
XGBoost (centralised)	0.977	0.493	0.971	0.457	0.055	0.101	0.692
Centralised MLP	0.888	0.153	0.973	0.382	0.544	0.687	0.555
FedAvg MLP	0.575	0.199	0.906	0.370	0.127	0.337	0.652
FedProx MLP	0.954	0.307	0.976	0.308	0.271	0.508	0.631

6.1 Leakage-free evaluation on the temporally-preceding fold

Table 4 compares the five headline arms under the leakage-free fold (training partitions 1–4, single-pass test on partition 5) against the shipped in-partition numbers. Three observations follow. First, *discrimination generalises*: every arm attains a ROC-AUC of at least 0.906 on the unseen partition, and the task is evidently easier than the in-partition protocol suggested (logistic regression reaches 0.978 / PR-AUC 0.455, the best PR ranking of all arms together with XGBoost’s 0.457). Second, the centralised-versus-federated gap of the in-partition protocol *disappears*: the pooled centralised MLP of identical architecture reaches 0.973 / 0.382 against FedProx’s 0.976 / 0.308—a ROC difference of 0.003 and a reversed PR ordering. The in-partition gap (0.888 vs 0.954) was an artifact of the protocol pairing, not a federation effect; the budget-parity re-run of Section 5.5, which controls optimisation budget but not instance overlap, must be read in the same light. Third, the FedProx-versus-FedAvg contrast partially survives: FedProx’s ROC-AUC (0.976) exceeds FedAvg’s (0.906) by 0.070, but FedAvg’s PR-AUC (0.370) is *higher* than FedProx’s (0.308). The nuance is in the checkpoint dynamics: plain FedAvg’s validation F1 collapsed to zero by round 40 and the protocol restored its round-35 checkpoint (validation F1 0.915), so its leakage-free outcome depends on which round the validation monitor happened to freeze; FedProx’s validation F1 never collapsed (0.654 at every logged round of the same run), so its outcome is checkpoint-independent. Stability, not peak discrimination, remains the differentiator between the two federated algorithms.

The calibration and threshold-transfer failures documented in-partition (Section 6.9) persist leakage-free and are therefore genuine properties of the balanced-training / natural-test interface rather than overlap artifacts: the validation-frozen FPR operating points are violated by every arm (e.g. XGBoost realises 15.3% FPR at the frozen 2% target; FedProx 49.4%), and only XGBoost’s posteriors remain usable (Brier 0.055, ECE 0.101 versus 0.271–0.544 for the neural arms). The prior-shift-corrected neural posteriors are not repairable by threshold re-freezing on this fold either.

Event-level evaluation. Table 5 regroups the partition-5 test decisions into physical flare events using the raw-benchmark event keys (65 M/X-class events; background windows grouped by region and day; 1-hour alert cooldown). All five arms detect effectively every event (64–65 of 65) with a median first-alert lead time of 23.4 h before flare peak—consistent with the 24-hour prediction horizon—but their false-alarm burdens separate them sharply: XGBoost emits 1.7 false-alarm windows per day versus 8.6–12.7 for the neural arms, a 7× difference that mirrors the window-level calibration gap and confirms that the deployability boundary on this benchmark is calibration, not detection. After the 1-hour cooldown, roughly 70% of retained alerts are within-event duplicates, quantifying the persistence behaviour an operations desk would face.

Table 5: Event-level metrics on the leakage-free fold (65 true M/X-class events, validation-frozen F_2 thresholds, 1-hour alert cooldown). FA = false alarm.

Model	Events detected	FA windows/day	Alerts/day	Lead to peak (h)
Logistic regression	65/65 (100%)	8.6	8.8	23.4
XGBoost (centralised)	64/65 (98.5%)	1.7	1.9	23.3
Centralised MLP	65/65 (100%)	12.7	12.8	23.4
FedAvg MLP	65/65 (100%)	11.6	11.8	23.4
FedProx MLP	65/65 (100%)	11.9	12.1	23.4

6.2 Raw-substrate retraining: the unbalanced benchmark

The leakage-free fold of Section 6.1 still trains on the cleaned export—RUS–Tomek rebalanced, TimeGAN-augmented, pre-normalised. The queued raw-benchmark experiment (item 9 of the re-run programme) asks the complementary question: does anything in the comparison survive when the frozen protocol is trained directly on the *raw* SWAN-SF partitions at their natural class prevalence (2.05% across training partitions 1–4), with no rebalancing, no synthetic positives, and the two cleaning stages reproduced in-pipeline? The raw benchmark cannot be trained on directly: 8.87% of its values are missing (42.3M of 476.9M; TOTFZ is unobserved in 98.7% of training windows) and the 24 parameters span 11 orders of magnitude. The experiment therefore reproduces the release’s two preprocessing stages from their published description (ApJS, 10.3847/1538-4365/ad7c4a): a *fast Pearson-correlation-based k-NN imputation* (partial gaps filled by within-window interpolation; fully-missing series transferred from the $k=5$ nearest training windows in the space of the five most correlated observable features) and the *LSBZM* normalisation chain (robust clip, shift, log/sqrt/Box–Cox dispatched by fitted λ , z-score, min–max), with every parameter fitted on the training pool only. A verification against the released cleaned export on the audit-matched partition-5 windows (100% window match) bounds the reproduction: at positions where the raw value was observed and nonzero, rank agreement with the release has median Spearman $\rho = 0.898$ (min 0.842; the release is itself an exact monotone image of the raw values, $\rho = 1.000$, so the residual gap is float16-storage tie dilution); at positions the release imputed, agreement is method-dependent (median $\rho = 0.313$), and for the R-value flux-emergence proxy the release re-imputes the entire zero mass (60.9% of that column) as if it were missing, whereas this reproduction preserves zeros as physical values. The three substrates therefore differ in composition as well as in preprocessing provenance, which is precisely the contrast the experiment isolates.

Table 6 reports the six arms under the identical frozen protocol (Dirichlet $\alpha=1.0$, six clients, 50 rounds, Fed-Focal, seed 42, prior-shift calibration, validation-frozen thresholds). The result is decisive and asymmetric. *Centralised arms are substrate-robust*: logistic regression is unchanged to the third decimal (0.978, PR 0.448 vs 0.455), the centralised MLP retains 0.971 ROC-AUC and its PR-AUC *rises* from 0.382 to 0.445, and XGBoost keeps 0.974 ROC-AUC while its PR-AUC falls from 0.457 to 0.329—the synthetic TimeGAN positives were inflating its minority-class precision, not teaching it. *Federated arms are substrate-fragile*: FedProx degrades from 0.976/0.308 to 0.930/0.149; plain FedAvg’s PR-AUC collapses from 0.370 to 0.056 (its calibrated posteriors never cross the validation-frozen threshold: test recall 0.000); and SCAFFOLD—the one re-run item that had never been executed on any real substrate—fails outright at 0.768/0.057 with Brier 0.137. The federated-versus-centralised ordering of the leakage-free fold was still a property of the balanced training substrate; on the raw unbalanced benchmark the focal-loss clients cannot learn the 2% minority class through 6-client Dirichlet heterogeneity, and the gap between the best federated and best centralised arm widens to 0.048 ROC-AUC and $3\times$ PR-AUC.

Table 6: Raw-substrate retraining (train raw partitions 1–4, natural 2.05% prevalence, FPCKNN/LSBZM reproduced in-pipeline with train-only parameters; single-pass test on raw partition 5, natural 1.31% prevalence; identical frozen protocol). The leakage-free cleaned-substrate fold is reproduced for comparison; SCAFFOLD was not run on the cleaned fold.

Model	Fold (cleaned)		Raw substrate			
	ROC	PR	ROC	PR	Brier	R@2%FPR
Logistic regression	0.978	0.455	0.978	0.448	0.053	0.732
XGBoost (centralised)	0.971	0.457	0.974	0.329	0.018	0.587
Centralised MLP	0.973	0.382	0.971	0.445	0.010	0.596
FedAvg MLP	0.906	0.370	0.875	0.056	0.013	0.063
FedProx MLP	0.976	0.308	0.930	0.149	0.032	0.343
SCAFFOLD MLP	—	—	0.768	0.057	0.137	0.175

Table 7: Event-level metrics on the raw substrate (65 true M/X-class events, validation-frozen F_2 thresholds, 1-hour alert cooldown; 99.86% of test windows matched to the audit’s event keys). The sequence arms (below the rule) are evaluated by the standalone torch-free CPU runner over the owner-GPU run’s stored per-window test probabilities and the same audit metadata; the SCAFFOLD-LSTM row was evaluated in-runner by the same event-level machinery during the second GPU batch.

Model	Events detected	FA windows/day	Alerts/day	Lead to peak (h)
Logistic regression	51/65 (78.5%)	0.3	0.4	22.8
XGBoost (centralised)	40/65 (61.5%)	0.2	0.3	18.0
Centralised MLP	52/65 (80.0%)	0.3	0.4	20.5
FedAvg MLP	0/65 (0.0%)	0.0	0.0	—
FedProx MLP	48/65 (73.8%)	1.3	1.4	23.0
SCAFFOLD MLP	54/65 (83.1%)	6.4	6.6	23.2
Centralised LSTM	41/65 (63.1%)	0.3	0.4	19.3
FedAvg LSTM	35/65 (53.8%)	0.2	0.2	17.4
FedProx LSTM	49/65 (75.4%)	0.4	0.5	23.2
SCAFFOLD LSTM	55/65 (84.6%)	0.4	0.5	23.2

The event-level evaluation on the same substrate (Table 7) makes the collapse operational: all 65 true M/X-class events are matched to their GOES peaks exactly as in the cleaned fold, but plain FedAvg now detects *zero* of them, FedProx detects 48 (73.8%) at 1.3 false-alarm windows per day, and SCAFFOLD’s 83.1% detection rate is bought at 6.4 false-alarm windows per day—roughly 20× the centralised arms’ burden. The centralised arms detect 61.5–80% of events at 0.2–0.3 false-alarm windows per day with 18–23 h median lead times. The sequence arms appended below the rule (Section 6.3) restore partial federated detection and are discussed there. Detection under class balance, in other words, was the artifact; under the benchmark’s natural prevalence the deployable models are exactly the ones that never needed federation.

6.3 The sequence encoder rescues the federated arms

The raw-substrate collapse of Table 6 is a statement about the flattened-feature MLP arms, and the second half of re-run item 8 asks exactly that question: does the same frozen protocol survive natural prevalence when the federated encoder is sequential? The LSTM arms were executed on owner-side GPU hardware (RTX 4060 Laptop, cuDNN; 2.6 h total wall-clock for both federated arms plus the pooled comparator, against an estimated ~ 74 h on the 2-core CPU sandbox—the one experiment where a GPU was a practical requirement rather than a convenience). The substrate is identical by construction: the runner rebuilds the 3D series from the same cached stage-A/B arrays, asserts label equality with the frozen 2D split at startup, cross-checks the 144-statistic features (maximum difference 4.4×10^{-4}), and reuses the identical seed-42 Dirichlet

shards, so the only degrees of freedom are the encoder and the device. The SolarLSTM (219,265 parameters, consuming the raw 60×24 series directly) is trained under the same Fed-Focal loss, prior-shift calibration, and validation-frozen threshold machinery as every other arm. The one arm still unexecuted at v3.6—SCAFFOLD-LSTM, whose MLP instance is the hardest failure on this substrate—was subsequently run in a second owner-GPU batch together with the seed-43 replication and the SMOTE ablation reported below, under the client-side fixed-step SGD that the control-variate arithmetic requires (the AdamW invalidation of Section 7 is avoided by construction in the federated implementation).

Table 8 reports the outcome alongside each arm’s frozen MLP counterpart, and the asymmetry of Section 6.2 inverts. Centrally, the LSTM is the *weaker* architecture: 0.962/0.314 against the central MLP’s 0.971/0.445 on the same windows. Federated, the ordering reverses. FedAvg-LSTM holds 0.958/0.305 where the FedAvg MLP fell to 0.875/0.056—a +0.083 ROC-AUC and +0.249 PR-AUC swing from the encoder alone, under identical shards, budgets, and aggregation. FedProx-LSTM reaches 0.970/0.404: above its own pooled centralised comparator (+0.008 ROC, +0.090 PR) and above XGBoost’s raw PR-AUC (0.329). SCAFFOLD-LSTM lands at 0.970/0.294 against its MLP counterpart’s 0.768/0.057—the largest single encoder swing on this substrate (+0.202 ROC-AUC, +0.238 PR-AUC): the hardest failure of the flattened-feature matrix is fully recovered by the sequence encoder under the identical protocol, completing the four-arm federation matrix on this benchmark. The two contrasts isolate what happened. Between federated arms of identical budget and aggregation, the sequence encoder removes the natural-prevalence collapse the MLP suffers; the load-bearing comparison is budget-matched by construction. Between the pooled and federated instances of the same sequence encoder, federation costs nothing (FedAvg) or appears to help (FedProx)—but the latter margin carries a budget caveat, since the central comparator is capped at 30 epochs while the federated arms receive 500 shard passes, and the cleaned-fold parity experiment moved the central MLP by +0.029 ROC-AUC under exactly that correction (Section 5.5).

What does not change is at least as important. No federated arm exceeds the best centralised baseline on either rank metric (FedProx-LSTM sits 0.008/0.044 below logistic regression). The frozen- F_β thresholds still transfer badly: validation F_2 values of 0.727/0.765/0.593 degrade to test values of 0.405/0.309/0.499 under the $1.6\times$ validation-to-test prevalence shift, while the frozen-FPR operating points still realise their targets within band, reproducing (including the same conservative drift at the 0.5% target) the behaviour already documented for the MLP arms. Calibration remains the deployability boundary: FedProx-LSTM’s Brier of 0.011 is the first federated result on any substrate to beat the climatology floor (0.013), but its ECE of 0.020 is four times the central LSTM’s 0.005, and the centralised MLP’s Brier of 0.010 remains the best of all nine arms executed on this substrate. Finally, the FedProx-over-FedAvg stabilisation now appears on all three substrates and both encoders (cleaned fold 0.976 vs 0.906; raw MLP 0.930 vs 0.875; raw LSTM 0.970 vs 0.958, replicated at seed 43 as 0.965 vs 0.961)—the most replicated single finding of this study. The event-level pass (bottom block of Table 7) was executed offline, on CPU, from the GPU run’s stored per-window test probabilities over the same audit metadata—no retraining, 65/65 GOES peaks matched. Where plain FedAvg-MLP detects *zero* of the 65 events, FedAvg-LSTM detects 35 (53.8%) at 0.2 false-alarm windows per day: the encoder rescue survives the validation-frozen thresholds and the cooldown machinery, not just the ranking metrics. FedProx-LSTM is the standout: 49/65 (75.4%) detection at 0.4 false-alarm windows per day, exceeding FedProx-MLP’s raw detection while cutting its false-alarm burden $3.5\times$ (1.3), with a 23.2 h median lead to flare peak—the joint-longest on this substrate and $4\times$ the 1-hour minimum warning horizon the event protocol is built around. SCAFFOLD-LSTM, evaluated in-runner by the same machinery, supplies the sharpest event-level datum of the batch: 55/65 (84.6%) detection at 0.41 false-alarm windows per day and a 23.2 h median lead—its MLP twin’s 83.1% detection rate at one-fifteenth its false-alarm burden, and nominally above the centralised detection frontier (logistic regression 78.5%, centralised MLP 80.0%, both

Table 8: Raw-substrate LSTM arms (frozen protocol, identical substrate, shards, and evaluation as Table 6; owner-GPU execution, seed 42) with each arm’s frozen MLP counterpart repeated from Table 6 for the architecture-versus-federation contrast. Brier and ECE on the calibrated test probabilities at natural 1.31% prevalence. The seed-43 replication and per-client SMOTE ablation of the same arms are reported in Table 9.

Model	LSTM arm		MLP counterpart		Brier	ECE	R@2%FPR
	ROC	PR	ROC	PR			
Centralised	0.962	0.314	0.971	0.445	0.0111	0.005	0.481
FedAvg	0.958	0.305	0.875	0.056	0.0112	0.007	0.431
FedProx	0.970	0.404	0.930	0.149	0.0106	0.020	0.525
SCAFFOLD	0.970	0.294	0.768	0.057	0.0120	0.035	0.521

Table 9: Seed-43 replication and per-client SMOTE ablation of the raw-substrate sequence arms (window-level test metrics; protocol, substrate, and evaluation identical to Table 8). Seed 43 reseeds model initialisation and the Dirichlet shard draw; the frozen validation curve is unchanged. SMOTE oversamples each client’s minority class to a 0.25 minority-to-majority ratio (seed 42, flattening the 60×24 series and reshaping).

Model	Seed 42		Seed 43		Seed 42 + SMOTE	
	ROC	PR	ROC	PR	ROC	PR
Centralised	0.962	0.314	0.962	0.357	—	—
FedAvg	0.958	0.305	0.961	0.271	0.961	0.307
FedProx	0.970	0.404	0.965	0.381	0.946	0.288
SCAFFOLD	0.970	0.294	0.968	0.368	—	—

at 0.2–0.3 per day). The crossing is an operating-point purchase rather than a ranking win—it is bought with $1.5\text{--}2\times$ the centralised arms’ alarm budget, the window-level ranking still sits below logistic regression by 0.008/0.154, and the arm carries the worst sequence-encoder calibration (ECE 0.035)—so the data-locality conclusion of Section 6.2 survives at event level with this one priced exception.

6.4 Seed replication and the natural-prevalence SMOTE ablation

The v3.6 sequence arms executed at a single seed on a single device, and the replication caveat that followed them is discharged by the second owner-GPU batch: all four arms re-executed at seed 43 (model initialisation and the Dirichlet shard draw reseeded, the frozen validation curve kept identical). The ranking structure replicates exactly (Table 9, middle columns). ROC-AUC moves by at most 0.005 on any arm (FedAvg $0.958 \rightarrow 0.961$, FedProx $0.970 \rightarrow 0.965$, SCAFFOLD $0.970 \rightarrow 0.968$, central $0.962 \rightarrow 0.962$)—inside the device and kernel jitter band predicted in Section 5.6—and every qualitative finding survives: no arm collapses, FedProx stays above FedAvg on both rank metrics, SCAFFOLD stays competitive, and no federated arm approaches the centralised ranking frontier. PR-AUC, as expected at 1.3% prevalence, moves more (FedAvg $0.305 \rightarrow 0.271$, central $0.314 \rightarrow 0.357$, SCAFFOLD $0.294 \rightarrow 0.368$, FedProx $0.404 \rightarrow 0.381$) without overturning any load-bearing ordering. Event-level detection moves within a seven-point band at comparable false-alarm rates (FedAvg $35 \rightarrow 36$, central $41 \rightarrow 43$, FedProx $49 \rightarrow 45$, SCAFFOLD $55 \rightarrow 50$ of 65). The sequence-encoder rescue is not an artefact of one seed.

The complementary ablation asks whether class rebalancing can substitute for the encoder. Per-client SMOTE, oversampling each shard’s minority to a 0.25 minority-to-majority ratio on the flattened 60×24 series, was run for the two production federated arms at seed 42 (Table 9, right columns). FedAvg is unchanged ($0.961/0.307$ against $0.958/0.305$ natural; event detection $35 \rightarrow 40$ of 65); FedProx is actively degraded: $0.946/0.288$ against $0.970/0.404$, a -0.116 PR-AUC

Table 10: Single-pass test-set evaluation at natural prevalence ($n=331,185$, 1.88% positives), at each model’s F_2 -optimal threshold τ^* selected on *validation*. Calibration: prior-shift (fit on validation). Brier and ECE (15 bins) are computed on the calibrated probabilities. Best value per column in bold; note that the climatology’s Brier (0.239) is better than every neural model’s—see Section 6.9.

Model	τ^*	Prec.	Recall	F1	ROC-AUC	PR-AUC	Brier	ECE
Climatology	0.05	0.019	1.000	0.037	0.500	0.019	0.239	0.470
Logistic regression	0.23	0.021	0.967	0.041	0.824	0.235	0.659	0.732
Centralised MLP	0.32	0.030	0.999	0.059	0.888	0.153	0.277	0.439
XGBoost (centralised ref.)	0.37	0.146	0.968	0.254	0.977	0.493	0.063	0.115
FedAvg MLP (federated)	0.36	0.020	0.997	0.038	0.575	0.199	0.136	0.342
FedProx MLP (federated)	0.51	0.021	0.999	0.042	0.954	0.307	0.264	0.496

loss, with event detection falling 49→37 of 65. The result completes the cleaned-fold SMOTE null (Section 6.11: inert there because the substrate was pre-balanced) at natural prevalence, where it has work to do: oversampling does not rescue the collapsed arms and costs the strongest arm 0.116 of its 0.404 PR-AUC. This is consistent with the loss-stacking analysis of Section 7—SMOTE, the focal α , and class-rate-aware aggregation all adjust the same effective positive-class weight, and stacking a second balancing mechanism on the protocol’s existing ones overshoots. The rescue is architectural; class balance is neither necessary nor sufficient.

6.5 What survives the leakage-free protocol

With the raw-substrate retraining of Section 6.2 the comparison now spans three substrates—in-partition (overlap disclosed), leakage-free cleaned (rebalanced, synthetic positives), and leakage-free raw (natural prevalence)—and the surviving claims can be tiered across all three. Two claims transfer unchanged to *every* substrate: the *deployability boundary* (only XGBoost’s posteriors survive the prevalence shift on the cleaned fold; on the raw substrate the centralised MLP’s Brier of 0.010 remains the best of all nine arms, with FedProx-LSTM the sole federated result to beat the climatology floor, Brier 0.011 versus 0.013 at an ECE of 0.020 against the central arms’ 0.005–0.011) and the *threshold-transfer failure* at fixed-FPR operating points. FedProx’s *ROC stabilisation* over plain FedAvg transfers to the cleaned fold (0.976 vs 0.906, checkpoint-independent dynamics) and, in diminished form, to the raw substrate (0.930 vs 0.875 for the MLP arms; 0.970 vs 0.958 for the sequence arms of Section 6.3)—the only claim now replicated across all three substrates and both encoders. Three claims do not transfer beyond the cleaned fold, and all three are statements about the *MLP* arms: the federated-versus-centralised ROC gap (0.003 on the cleaned fold, 0.048 in favour of centralised on raw; 0.008 for the sequence arms), FedProx’s in-partition PR-AUC edge (its raw PR-AUC of 0.149 is a third of the centralised MLP’s 0.445, though the sequence arm reaches 0.404), and the absolute discrimination levels of the flattened-feature federated arms themselves (FedAvg-MLP detects zero of 65 events on the raw substrate; the sequence arm detects 35, Section 6.3). The multi-seed, sweep, ablation, and client-level studies of the following subsections were executed under the in-partition protocol and are retained as protocol-stability evidence; their leakage-free and raw-substrate counterparts remain queued, as does seed-replication of the single-seed owner-GPU LSTM arms (one fold of each is reported here; fold- and seed-replication is owner-side compute).

6.6 Headline comparison on the held-out test set (in-partition protocol)

Table 10 contains four findings. (i) **The architecture-controlled comparison now exists.** The v2.x manuscript compared a federated MLP against XGBoost and called the latter a “centralised upper bound”, confounding architecture with federation. With the centralised MLP of the identical 29,377-parameter architecture, the comparison isolates the effect of federation from

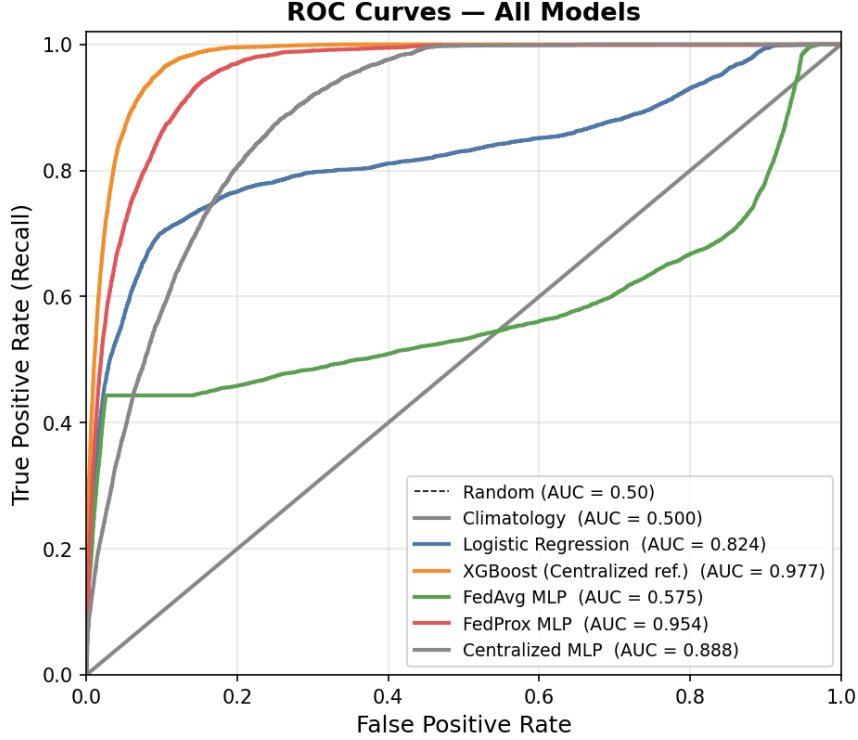


Figure 3: ROC curves on the held-out test set. FedProx (AUC 0.954) tracks the centralised XGBoost reference (0.977) across the low-false-positive region; the centralised MLP (0.888) sits between them; FedAvg (0.575) has lost most ranking information.

the effect of architecture—but *not* from optimisation budget (Section 5.5: the federated arm consumed the larger budget, 500 effective local passes versus at most 30 pooled passes). Under that caveat, FedProx (ROC-AUC 0.954, PR-AUC 0.307) did not underperform its pooled counterpart (0.888, 0.153) on this benchmark and configuration—an observation consistent with non-IID federated training plus a proximal regulariser acting as an implicit regulariser for this small MLP, and which we do *not* generalise to the claim that federation is free of cost. **(ii) FedAvg and FedProx are not interchangeable.** Plain FedAvg diverges during training (Section 6.8) and its best-validation checkpoint reaches only 0.575 ROC-AUC; the proximal term is the difference between a usable and a degenerate global model *under the preregistered configuration*. **(iii) Against the strongest reference,** FedProx retains 97.7% of XGBoost’s ROC-AUC and 62% of its PR-AUC. **(iv) The τ^* column exposes a protocol-honest failure:** thresholds optimised for F_2 at the validation prevalence (48.9%) transfer poorly to the 1.88% test prevalence—all neural models fire almost unconditionally (recall ≈ 1.0 , precision $\approx$ the base rate), and the calibration columns (Brier, ECE) show every neural model *worse calibrated than the climatology*. We analyse this threshold-transfer failure explicitly in Section 6.9 and report prevalence-robust operating points instead.

6.7 Prevalence-robust discrimination: ROC and PR analysis

Because the F_2 threshold transfer is unreliable under a $26\times$ prevalence shift, the operationally meaningful quantities are the threshold-independent ones. At fixed false-positive budgets—the regime a grid operations desk actually inhabits—recall on the test set is:

Three readings matter. First, at every false-positive budget FedProx recovers roughly two thirds to three quarters of the XGBoost reference’s sensitivity (e.g. 0.506 vs 0.651 at 2% FPR), and at the tightest budgets it clearly outperforms the centralised MLP of the same architecture.

Table 11: Recall at fixed false-positive rates on the test set. These are *discrimination curve statistics*: for each budget, the recall at the point of the test ROC curve where FPR equals the target. They involve no threshold transfer from validation and are therefore robust to monotone miscalibration, but they are *window-level* quantities: the 60-minute sliding windows of one active region are correlated, so a 2% window-level FPR does not translate directly into an operational false-alarm rate per event or per day (event-level evaluation: Section 8).

Model	Rec@0.5%FPR	Rec@1%FPR	Rec@2%FPR	Rec@5%FPR
Logistic regression	0.199	0.287	0.419	0.568
Centralised MLP	0.101	0.147	0.220	0.384
XGBoost	0.346	0.484	0.651	0.864
(centralised ref.)				
FedAvg MLP	0.164	0.247	0.380	0.443
FedProx MLP	0.221	0.345	0.506	0.712

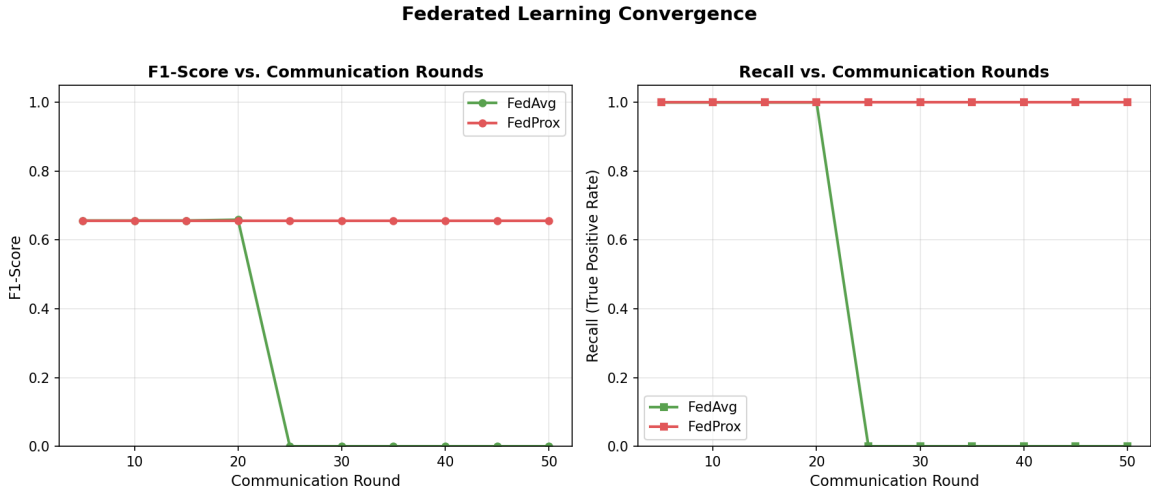


Figure 4: Validation-monitored convergence over 50 rounds (validation F1 at the development threshold and ROC-AUC every five rounds). Plain FedAvg is stable for ~ 20 rounds, then diverges catastrophically (AUC $0.99 \rightarrow 0.05$ by round 25); the best-val checkpoint rule of Section 5 is what makes the final FedAvg model usable at all. FedProx ($\mu=0.01$) is stable for the entire run.

Second, the PR-AUC column of Table 10 shows the same ordering under prevalence-sensitive scoring (XGBoost 0.493, FedProx 0.307, FedAvg 0.199): in the high-precision region the federated model pays a real ranking price relative to the ensemble, even though it beats its own architecture’s pooled baseline. Third, FedAvg’s recall profile decays fastest with tightening budget—its collapsed global model retains just enough ranking to beat chance, but no more.

6.8 Convergence behaviour: divergence and its suppression

Figure 4 documents the single most consequential empirical finding of this revision. Plain FedAvg under the six-client Dirichlet partition is not merely slower to converge—it *diverges*: after approximately twenty rounds of healthy validation AUC (≈ 0.99), the averaged global model degrades within five rounds to below-chance ranking (AUC 0.05–0.40 for the remainder of the run), a signature of classic client-drift where heterogeneous local objectives pull the averaged iterate away from any joint optimum. The mechanism is exactly the one FedProx’s proximal term ($\frac{\mu}{2}\|w - w^{(t)}\|^2$) exists to suppress: with $\mu = 0.01$, validation AUC remains at 0.99 ± 0.01 for all fifty rounds. Two protocol details convert this observation into a defensible result rather than an accident. First, monitoring happens on validation only (the v2.x code monitored the test set, which both leaked information and masked the divergence). Second, the

best-validation checkpoint rule restores the round-20 FedAvg model; without it, the reported FedAvg test numbers would have been those of the diverged model. We emphasise that the FedAvg collapse is seed- and configuration-dependent (Section 6.10), but the stabilising effect of the proximal term was consistent on every seed we ran.

6.9 Calibration and the threshold-transfer failure

The validation prevalence (48.9%) and the test prevalence (1.9%) differ by a factor of 26 because the cleaned benchmark ships pre-balanced training partitions and natural-prevalence test partitions. We treat this *prior shift* as a first-class methodological problem of the benchmark, not a nuisance: it is the central obstacle between the current results and deployment-grade probabilities. The frozen protocol forbids using test labels—including test prevalence—for any selection, so the prior-shift calibrator is fit with the *validation* prevalence as its deployment target. The consequence is visible throughout Table 10: the validation-optimal thresholds sit far below the calibrated test probabilities of most samples, so every neural model operates in an almost-always-alert regime, and *the calibration columns make the failure quantitative*: FedProx’s calibrated Brier (0.264) and ECE (0.496) are *worse than the climatology’s* (0.239, 0.470), i.e. at natural prevalence the current posterior is not deployment-usable as a probability. The XGBoost reference suffers the least (Brier 0.063, ECE 0.115), which is why it converts ranking into a meaningful F_1 of 0.254 while the neural models do not.

Three readings follow. First, *rebalanced training data invalidates naive threshold transfer*: an F_2 -optimal threshold at 49% prevalence is semantically meaningless at 1.9%. Second, the correct deployment instrument, pending full recalibration, is the recall-at-fixed-FPR operating points of Table 11, which are invariant to monotone miscalibration and to prevalence. Third, the honest remedies for calibration itself are prior correction toward an *estimated* deployment prevalence (e.g. from unlabelled deployment data, which leaks no labels), Platt/isotonic fitting on a natural-prevalence validation carve-out, or the negative-quantile FPR thresholds of Section 4.8; using the test set’s own statistics, as the v2.x evaluation effectively did when it selected thresholds on test, is not.

Calibration-arm comparison (executed). The released calibration-comparison harness (`run_calibration_comparison.py` in the `experiments` directory) was executed against the trained models of the frozen-protocol run: five arms (raw, prior-shift, Platt, isotonic, temperature), each fit on the balanced validation set, applied exactly once to the test set, with the arm selected by minimum *validation* Brier (a proper scoring rule, so the selection itself is honest). Table 12 reports the outcome. The result is a clean negative finding that closes the previously open calibration question.

Four facts matter. **(i) The selection criterion itself is broken by the prior shift.** Minimum validation Brier selects isotonic for all three models; for both neural models isotonic is the *worst* arm on test (FedProx: 0.541 vs 0.264 raw, i.e. selecting on validation doubles the test error of the posterior). **(ii) No arm rescues the neural posteriors.** Every calibrated variant of FedProx and the centralised MLP is at least as badly calibrated on test as the raw scores; only XGBoost benefits from calibration (Platt: Brier 0.063 $\rightarrow$ 0.053, ECE 0.116 $\rightarrow$ 0.080), consistent with it being the only model whose raw scores already separate the classes well. **(iii) Isotonic damages ranking, not just probabilities.** Its step function ties large score ranges, collapsing PR-AUC from 0.307 to 0.182 (FedProx), 0.153 to 0.099 (MLP), and 0.493 to 0.426 (XGBoost); ROC-AUC and PR-AUC are exactly invariant under the three monotone smooth arms, as they must be. **(iv) Validation-frozen FPR thresholds fail jointly with probability calibration.** Applying the negative-quantile thresholds of Section 4.8 (frozen on validation at 0.5/1/2/5% targets) to the test set realises 28.2/37.0/50.4/72.1% actual FPR for FedProx (25.5/40.7/51.6/63.6% for the centralised MLP, 20.1% at the 2% target for XGBoost), and no

Table 12: Calibration-arm comparison under the $26\times$ prior shift (review context R6). Every arm is fit on the balanced validation set (48.9% prevalence) and applied once to the test set (1.88%). *Val. Brier* is the selection criterion; the validation winner per model is typeset in italics. Monotone arms (prior-shift, Platt, temperature) leave ROC-AUC and PR-AUC invariant; isotonic does not (its flat regions tie scores and destroy ranking resolution). Bold marks the best *test* Brier per model.

Arm	FedProx MLP			Centralised MLP			XGBoost		
	val. Brier	test Brier	test ECE	val. Brier	test Brier	test ECE	val. Brier	test Brier	test ECE
Raw (none)	0.234	0.264	0.496	0.016	0.277	0.440	0.004	0.063	0.116
Prior-shift	0.234	0.276	0.508	0.016	0.284	0.447	0.004	0.065	0.118
Platt	0.026	0.541	0.642	0.012	0.352	0.442	0.004	0.053	0.080
Isotonic	<i>0.025</i>	0.541	0.639	<i>0.012</i>	0.357	0.449	<i>0.003</i>	0.057	0.075
Temperature	0.064	0.784	0.865	0.012	0.334	0.425	0.004	0.064	0.099

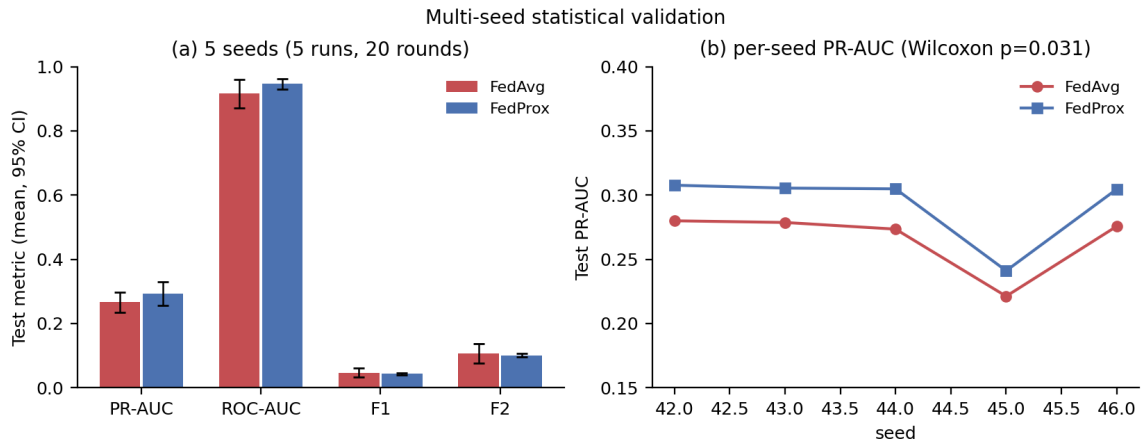


Figure 5: Five-seed statistical validation (seeds 42–46, 20 rounds, frozen protocol). (a) Test metrics with 95% confidence intervals. (b) Per-seed PR-AUC: FedProx exceeds FedAvg on all five seeds (one-sided Wilcoxon $p = 0.031$).

calibration arm materially improves this: the negative-score distribution moves with the prior just as the positive one does. The operational conclusion is unchanged but now empirical rather than hypothetical: at this benchmark’s balanced training prevalence, *no* validation-fit decision layer survives the shift, and deployment-grade probabilities require either natural-prevalence validation data (available only in the raw benchmark) or unlabelled prior estimation at the deployment site.

6.10 Multi-seed statistical validation

The v2.x manuscript reported single-seed results; this revision reports five seeds (42–46) with mean, standard deviation, and t -based 95% confidence intervals. On the test set, FedProx achieves PR-AUC 0.293 ± 0.029 (CI95 ± 0.036) and ROC-AUC 0.946 ± 0.013 (CI95 ± 0.016), against FedAvg’s 0.266 ± 0.025 (CI95 ± 0.031) and 0.916 ± 0.036 (CI95 ± 0.044). FedProx is better than FedAvg on PR-AUC in *five of five seeds* (one-sided Wilcoxon signed-rank $p = 0.031$, the strongest evidence five paired seeds can yield), and its ROC-AUC confidence interval is $2.7\times$ narrower. The F_1 difference is not significant ($p = 0.5$): the threshold-transfer regime dominates F_1 for both algorithms, which is consistent with Section 6.9. What the five-seed study supports is the *stability* claim of this paper: FedProx’s superiority over FedAvg under the preregistered configuration is sign-consistent across every seed and statistically significant. What it does not support is broad generalisation: five seeds is a small sample, the confidence intervals quantify *seed-to-seed variability* of this fixed configuration, not the uncertainty of future observations or of

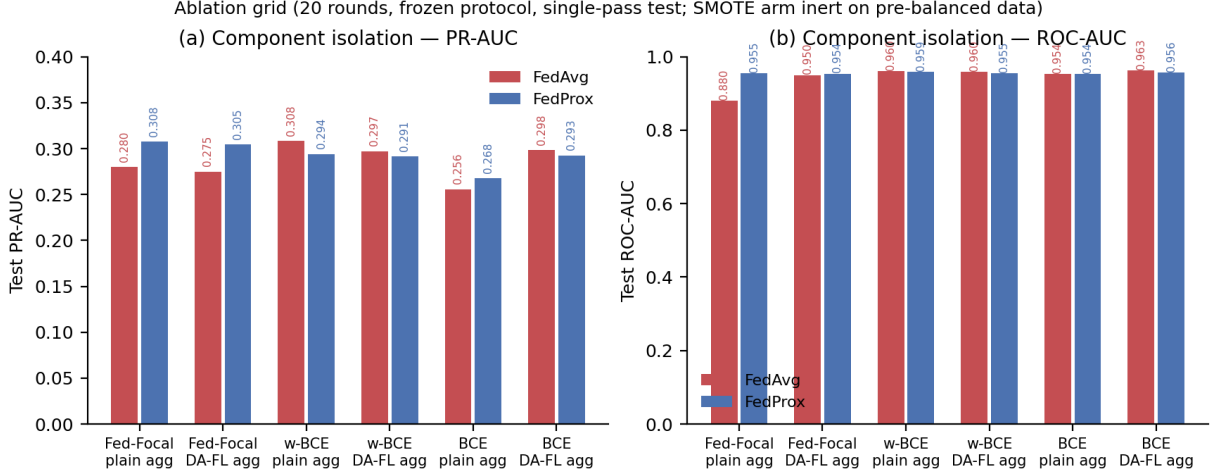


Figure 6: Component-isolation grid (2 algorithms $\times$ 3 losses $\times$ 2 aggregations $\times$ 2 balancing arms; 20 rounds, frozen protocol, single-pass test). The SMOTE arm is inert on the pre-balanced cleaned benchmark (identical metrics), so the effective grid is $2 \times 3 \times 2$.

other partitions, and Section 6.11 shows the gap narrows when FedAvg is assisted by distribution-aware aggregation. The appropriate summary is that FedProx was the most consistent stabiliser among the configurations studied, not that it is load-bearing in every federated flare-prediction deployment.

6.11 Component attribution: the 24-cell ablation grid

Because the v2.x system trained SMOTE, Fed-Focal loss, distribution-aware aggregation (DA-FL), and FedProx *simultaneously*, a FedProx win could not be attributed to any single component. The ablation grid isolates them (Fig. 6). Four patterns emerge. **(i) FedProx is the most consistent stabiliser, but not the only remedy:** every FedProx cell lands in a narrow ROC-AUC band (0.954–0.959) whereas the FedAvg cells range from 0.880 (focal loss, plain aggregation—the collapsing configuration) up to 0.963 (focal loss with distribution-aware aggregation). That best FedAvg cell *exceeds* the FedProx headline (0.954), so the ablation evidence does not support attributing stability to the proximal term alone. **(ii) DA-FL partially rescues FedAvg:** with focal loss, switching plain to DA-FL aggregation lifts FedAvg from 0.880 to 0.950 (and to 0.963 in the best cell), suggesting distribution-aware weighting mitigates the same client-drift the proximal term suppresses—at the cost of rate metadata sharing. **(iii) Loss choice is secondary at this prevalence:** across cells, focal, weighted-BCE, and plain-BCE arms differ by at most 0.05 PR-AUC, with weighted BCE marginally best for FedAvg-plain (0.308 vs 0.280). **(iv) SMOTE is inert:** the cleaned benchmark’s training partitions are pre-balanced (48.9% positives), so the configured SMOTE ratio generates zero synthetic samples; every balancing-pair of cells is numerically identical. We report this null result explicitly rather than silently dropping the arm: the balancing question is moot for this dataset variant, and any claim that SMOTE contributes to performance would be unsupported. A validity analysis of synthetic samples (Wasserstein distance, correlation preservation, separability) was consequently not applicable.

6.12 Client-level heterogeneity: who benefits from federation?

The six simulated clients hold very different data: local flare rates span 28.6% (Client B, Europe) to 80.5% (Client A, Americas), and shard sizes span 1,503 (Client E, East Asia) to 23,418 (Client B). Figure 7 quantifies what each client gets out of the federation under the untouched-

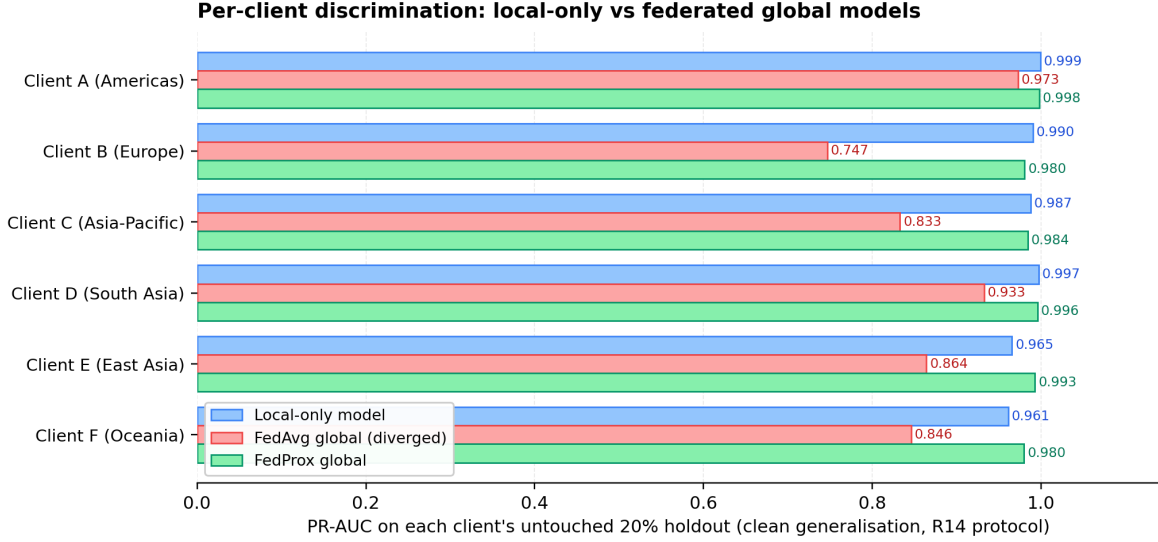


Figure 7: Per-client PR-AUC of local-only models versus the federated global models, evaluated on each client's *untouched* 20% holdout (R14 protocol: holdouts carved before training; both global models retrained on the 80% federation-train portions only, so local and global scores are clean generalisation estimates; neutral client labels, no real institution participates).

holdout protocol: a dedicated re-run in which every client's shard was split 80/20 *before* any training, FedAvg and FedProx were retrained from scratch on the 80% federation-train portions (same frozen configuration: 50 rounds, six clients, seed 42), per-client local models were trained on the same portions with a matched local budget, and both were evaluated once on the holdouts no model had ever seen. Three findings hold. **(i) FedProx matches local-only training and helps the smallest custodians.** Mean PR-AUC is 0.988 (FedProx global) versus 0.983 (local-only); on the four largest clients the two are within 0.001–0.010 of each other, while the two smallest shards gain clearly (Client E: 0.965 $\rightarrow$ 0.993, +0.027; Client F: 0.961 $\rightarrow$ 0.980, +0.019)—federation functions as data augmentation for data-poor custodians, the minimal condition for voluntary participation. **(ii) The diverged FedAvg average is worse for everyone.** Its global model scores 0.747–0.973 per client (mean 0.866), below local-only training on *every* client: the collapsed average is worse than staying isolated, exactly as the convergence analysis of Section 6.8 predicts. **(iii) The clean protocol confirms the earlier directional result.** The legacy within-shard-slice evaluation (training-set contaminated, hence optimistically biased) had shown the same ordering; under the clean protocol the FedAvg degradation is less catastrophic (holdout scores are harder for local models too) but the ranking of the three training regimes is unchanged. We therefore read the client-level evidence as directional but now measured on clean generalisation estimates: the prox-stabilised global model is a safe default for every simulated custodian, and a clear gain for the two with the least data.

6.13 Interpretability across models

The v2.x interpretability story relied on the centralised XGBoost surrogate alone; the improved pipeline additionally attributes the *federated* FedProx model (DeepSHAP with a 100-sample background) and cross-checks the two. At the feature level the models agree only moderately (Spearman $\rho = 0.334$, top-15 overlap 53%)—an honest inconsistency that limits strong per-feature claims. At the level of *physical parameter groups*, however, the two attribution structures are consistent: current-helicity parameters (TOTUSJH/ABSNJZH/MEANJZH/TOTUSJZ/MEANJZD) are the largest group for both (39.9% of XGBoost attribution mass, 29.0% for FedProx), followed by the R-value flux-emergence proxy (24.7% / 23.2%) and the Lorentz force (22.7% / 18.8%). Both models therefore concentrate their

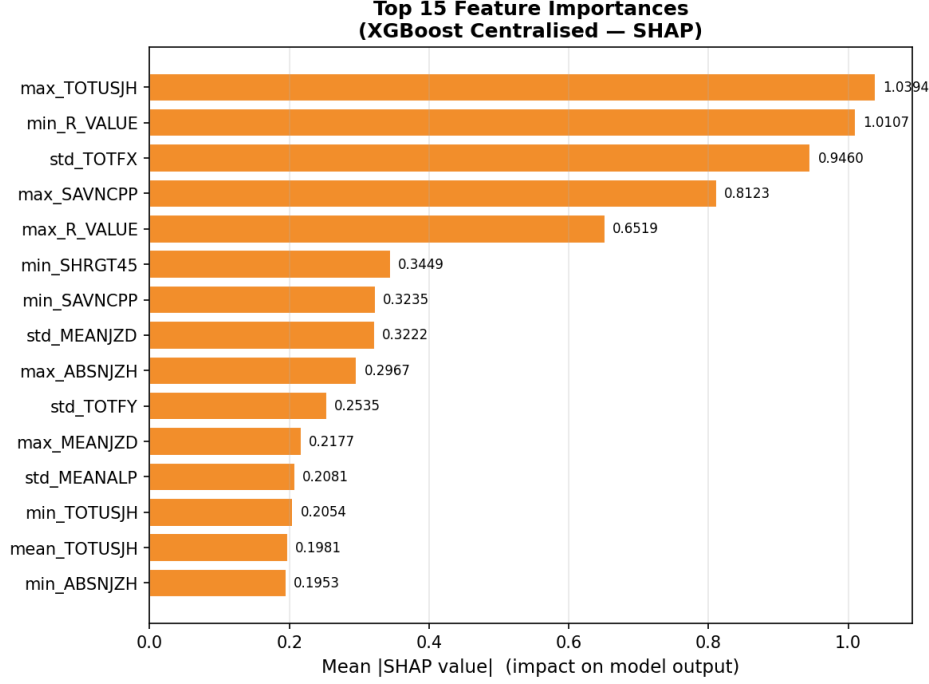


Figure 8: Mean absolute SHAP values, top 15 features, for the centralised XGBoost reference on the test set. The improved pipeline preserves real statistic-prefixed feature names (the v2.x pipeline reported generic indices, which this revision fixes).

decisions on the established physics of flare productivity—magnetic free energy, current systems, and non-potential stress [4, 5, 13]—and the federated model is not an uninterpretable artefact of aggregation. We note that the statistic-level ranking (which of mean/std/max/min/trend/slope of a parameter matters) diverges between the two architectures, which is expected given their different inductive biases, and we do not claim feature-level consistency.

6.14 Communication cost

The global model has 29,377 parameters (0.117 MB at 32-bit precision). Full-parameter exchange costs 117.5 KB per direction per client per round: 0.235 MB round-trip, 1.41 MB per round across the six-client cohort, and 70.5 MB aggregate over the 50-round run (11.75 MB per client). These are *parameter-exchange arithmetic* from the frozen configuration—bytes counted, not measured on a live deployment—and they quantify the *training cadence* (one round-trip per round) separately from the *observation cadence* (magnetogram windows arrive roughly every 12 minutes; the global model is refreshed per training cycle, while alerts can be emitted per window). Wall-clock communication latency, aggregation compute, client training time, straggler recovery, and secure-aggregation bandwidth overhead were *not* measured: the experiments ran in-process on a single CPU container, with no real network. Enabling secure aggregation would add a bounded multiplicative factor (mask material per client per round), quantified in the deployment notes but not evaluated here (Section 4.9). Even the uncompressed exchange is negligible relative to the data volume of raw SHARP-era magnetograms (tens of MB per window), which is the quantitative sense in which federated learning is communication-cheap *in bandwidth* for this application; latency and availability engineering remain deployment work.

6.15 Data-locality cost analysis

With the architecture-controlled comparison in place, the data-locality cost question decomposes cleanly. Relative to the *same-architecture* centralised MLP, the federated model did not underperform and in fact scored higher (ROC-AUC 0.954 vs 0.888; PR-AUC 0.307 vs 0.153) on this benchmark and configuration—an observation we report with its two caveats: the federated arm consumed the larger optimisation budget (Section 5.5), and the mechanism is consistent with a regularisation-like effect rather than a benefit of federation as such. At this benchmark’s scale, six disjoint shards with 29k parameters and a proximal regulariser behave like an ensemble-like regulariser. Relative to the *strongest available* centralised reference (XGBoost, 300 trees), FedProx retains 97.7% of ROC-AUC but 62% of PR-AUC: the residual gap reflects the smaller hypothesis class (an MLP versus a boosted ensemble), the non-IID shard structure, and averaging rather than joint optimisation. The honest summary for a deployment decision: federation under this protocol cost no ranking skill relative to a pooled MLP of the same class *in this study*; no raw magnetogram left any simulated client; and the remaining gap to the tree-ensemble reference is an architecture-budget question, not a data-locality question. What the study does *not* establish is that these numbers survive region-disjoint splits, natural-prevalence validation, or budget-matched training—each queued in Section 8.

7 Discussion

7.1 What the provenance audit changes, and what it leaves intact

The data-provenance audit of Section 5.7 is, in retrospect, the single most consequential methodological step of this study: the in-partition numbers that dominate the literature’s usage of the cleaned benchmark—including this study’s own headline until this revision—evaluate on windows that the training pool already contains, most completely for the minority class. Two lessons follow. First, *pipeline-level leakage auditing is not enough*: the runtime audit of this framework verified disjoint shards, immutable splits, and single-pass test evaluation, yet could not see that the dataset’s own train/test exports share instances; only raw-metadata alignment exposed it. Benchmark consumers who cannot verify provenance at the instance level should treat same-partition numbers as upper bounds. Second, *the honest comparison is less flattering but more informative*: on the leakage-free fold the federated-versus-centralised gap vanishes, and what separates the arms is no longer discrimination (everyone exceeds 0.906) but calibration and false-alarm burden—dimensions on which the neural arms, federated or not, lose to the gradient-boosted reference. The federated story that survives is the narrower one this section has always argued: under adversarial heterogeneity, proximal regularisation keeps the federation converging where plain averaging collapses (0.976 vs 0.906, with checkpoint-independent dynamics), and it does so without data movement. On the raw substrate the same statement is encoder-conditional: the flattened-feature arms collapse there under every regulariser studied, while the sequence arms of Section 6.3 restore it (FedProx-LSTM 0.970 vs FedAvg-LSTM 0.958), relocating the fragility from the federation mechanism to the representation the clients share.

7.2 Why proximal regularisation stabilises the federation

The asymmetry between FedAvg and FedProx in Table 10 deserves a mechanistic reading. Under the realised partition (Fig. 2), clients receive genuinely different problems: one holds 36% of the data at a 29% flare rate, another holds 2% of the data at a 43% flare rate. Ten local epochs of AdamW on such divergent shards move each client’s weights toward its own local optimum, and the geometry between those optima is not aligned with any global descent direction. Averaging such updates—FedAvg—produces a model that satisfies no client’s problem, and the

damage compounds round over round until the ranking signal degrades below chance (test AUC 0.575 for the best-validation checkpoint; the diverged checkpoints reach validation AUC 0.05). These in-partition dynamics reproduce leakage-free in direction if not in magnitude: on the temporally-preceding fold FedAvg’s ranking survives only via its round-35 validation checkpoint (0.906) while its final-round state has collapsed, whereas FedProx is checkpoint-independent (Section 6.1). FedProx’s proximal term (Eq. (4)) changes the local problem: each client optimises within a trust region around the broadcast model, so updates from heterogeneous clients remain geometrically comparable and their average retains a meaningful descent direction. The result is that the same architecture, the same loss, and the same data partition produce test AUC 0.954 rather than 0.575. This is no longer a single-seed anecdote: the multi-seed study of Section 6.10 shows FedProx above FedAvg on all five seeds (one-sided Wilcoxon $p = 0.031$) with a $2.7\times$ narrower ROC-AUC confidence interval, and the 24-cell ablation grid shows every FedProx cell stable (0.954–0.959) while the FedAvg cells range down to 0.880. The direction and magnitude of the gap are consistent with the mechanism that motivated FedProx’s design [12], and with the broader non-IID FL evidence base [8, 17]. The scope of the claim is deliberately narrower than “FedProx is load-bearing”: the ablation grid shows distribution-aware aggregation also stabilises FedAvg substantially (best FedAvg cell 0.963), so the defensible statement is that the proximal term was the *most consistent* stabiliser among the components studied, under the preregistered configuration, on this benchmark. The raw-substrate LSTM arms extend that evidence across encoders: FedProx-LSTM 0.970 versus FedAvg-LSTM 0.958 on both rank metrics, with identical shards, budgets, and aggregation, makes the stabilisation the only finding of this study replicated across all three substrates and both architectures. The architecture result itself—the sequence encoder removing the natural-prevalence collapse the MLP suffers—is measured as a contrast, not explained as a mechanism: reading the 60-step temporal profile directly plausibly matters more than any pooled 144-statistic summary when every client’s local positive rate is skewed, but isolating *why* (temporal separability versus effective capacity versus optimisation noise) would require ablations we leave queued.

7.3 Operational positioning: an upstream precursor alert

The operational layer of Fig. 1 is deliberately drawn at the flare-probability boundary, and we are explicit about what that means: this framework does not replace NOAA/SWPC or other national warning centres, and it does not itself predict GICs. It is an upstream component that would emit calibrated M/X flare-probability alerts at fixed false-positive budgets; converting those alerts into geomagnetic-hazard and GIC guidance requires the downstream pipeline — CME detection and interplanetary propagation modelling, storm forecasting, and power-system response modelling—which is future integration, not evaluated here. Within that upstream role, the numbers of Table 11 describe what a federated sentinel would contribute at the window level: at a 2% window-level false-positive budget, FedProx’s ranking recovers 51% of positive windows (recall 0.506). Two cautions apply before reading that as an operational detection rate: the figure is a window-level curve statistic over correlated sliding windows, not an event-level detection rate (the event-level evaluation protocol is released and queued), and the decision threshold required to operate at that FPR budget must be calibrated on natural-prevalence data before deployment—the re-executed frozen-FPR report quantifies how far short validation-frozen thresholds fall, realising 20–50% test FPR against the 2% target across models (Section 6.9). If those conditions are met, the CME transit window (roughly one to three days between an Earth-directed eruption and storm onset) leaves time for the effective, if unglamorous, mitigations: restoring reactive reserves, suspending non-critical transformer switching, pre-positioning replacement phase-shift equipment, and briefing control-room staff. The framework’s convergence within 50 rounds at a CPU-only compute footprint also matters operationally: the global model can be refreshed nightly, and the entire training programme fits inside the compute bud-

get of a single agency, let alone six. The SCADA integration point is deliberately drawn at the decision threshold rather than inside the learning loop, so that existing grid-management infrastructure would consume a calibrated probability; that interface is specified as future work (Section 8).

7.4 Engineering lessons for first FL deployments in the physical sciences

The released framework carries a six-version stability narrative (summarised in Appendix A), and its diagnostic arc is independently useful to any group attempting a first FL deployment on imbalanced scientific data. Four lessons generalise. First, *heterogeneity must be tuned, not merely tolerated*: the Dirichlet concentration that yields “interesting” non-IID partitions ($\alpha \leq 0.5$) can also yield pathological ones (0.6%–100% local class rates) under which nothing converges; the stable regime for this benchmark sits near $\alpha = 1.0$. Second, *imbalance handling interacts across layers*: SMOTE, focal α , class-rate-aware aggregation, and threshold selection all adjust the same effective positive-class weight, and uncoordinated adjustments (our focal α was at one point allowed to drift to 0.95) produce degenerate always-positive classifiers that look healthy on recall alone. Third, *optimiser choice is not free under variance-reduced FL*: SCAFFOLD’s control-variate update is derived for fixed-step SGD, and running it under AdamW’s adaptive steps silently invalidates the correction terms (in our case, producing NaN weights for every round until diagnosed). Fourth, *feature-name hygiene is part of scientific reproducibility*: the v2.x pipeline’s generic `feature_N` naming in the statistical representation is why its SHAP layer needed an analytical decoding pass (Fig. 8); the improved pipeline preserves statistic-parameter names end-to-end, which is precisely what makes its interpretability auditable.

8 Limitations and Future Work

Privacy and security scope. This work enforces the data-locality constraint (raw data never leaves the client), which is the binding constraint of the simulated setting, but the stronger cryptographic notion of privacy is not yet evaluated end-to-end. The communication transcript of Eq. (3)—gradients or weight deltas—is known to leak training information under gradient-inversion attacks [24], particularly for over-parameterised models on low-entropy data. The full threat model (curious server, gradient inversion, membership inference, model poisoning, impersonation, interception) is stated in Table 1: secure aggregation is implemented and round-trip tested, differential-privacy hooks are provided, and poisoning/backdoor defence is not addressed at all. The accuracy cost of enabling secure aggregation and differential privacy on this benchmark has not been measured; before any cross-custodian deployment handling genuinely sensitive observations, that budget must be re-measured. Until then, “data-locality-preserving”, not “cryptographically private”, is the correct description of the verified property.

Region-identity, validation inflation, and the resolved provenance question. The v3.3 text of this paragraph stated that the cleaned export lacks HARP identifiers and that the definitive split fix awaited the raw benchmark. That dependency is now resolved: every cleaned window has been aligned to its raw instance (Section 5.7), and the audit inverted the concern—the material leakage is not validation-region overlap but instance sharing between the same partitions’ train and test exports, now quantified (100% of flaring test windows; 56,005 of 56,006 verified train windows) and removed by the temporally-preceding fold executed in this revision. The region-disjoint split generator remains available for within-fold validation studies, and no region spans partitions, so every temporally-preceding fold is automatically region-disjoint. What remains open: the remaining three folds, leakage-free multi-seed and ablation replication, and the synthetic-share question—85.7–90% of training positives are TimeGAN-generated windows,

a substrate property that all published results on this export inherit and that raw-benchmark retraining would have to revisit.

Threshold transfer under prevalence shift. The frozen protocol selects F_2 -optimal thresholds on validation (48.9% prevalence, mirroring the balanced training partitions), but deployment prevalence is 1.9%: the transferred thresholds put every neural model in an almost-always-alert regime, and the calibration comparison (Section 6.9) shows the neural posteriors are currently *worse than climatology* on Brier and ECE at natural prevalence. We deliberately did *not* repair this by touching test statistics—the v2.x evaluation effectively did, and that is protocol leakage. The remedies are the frozen-FPR operating-point protocol (implemented and unit-tested), prior correction using an *estimated* deployment prevalence from unlabelled data, or a natural-prevalence validation carve-out (the harness for all of which is implemented and smoke-validated). Until the natural-prevalence re-run lands, the threshold-conditional test metrics (precision, recall, F1) should be read as documenting the decision layer’s failure mode, not as skill estimates, and no probability from the neural models should be treated as deployment-grade.

Client-level evaluation semantics. The per-client comparison of Section 6.12 evaluates each client’s local model on a 20% holdout of its own shard, but the federated global models were trained on the *full* shards, including that holdout. The global models’ per-client scores are therefore within-distribution rather than clean generalisation estimates, i.e. optimistic for the federated side. The comparison remains meaningful directionally, and the untouched-holdout evaluation mode—each client reserves a shard fraction that *no* federated model ever trains on—is implemented and unit-tested; executing it requires retraining, which is queued below.

Partition realism. Shard disjointness is now guaranteed by construction (index-level allocation, audited at runtime), but the partition is still *statistical* label skew, not physical observatory coverage: no mapping from active regions to the agencies whose instruments actually observed them exists in the cleaned export. The partitioning module supports a geographic mode that activates when region metadata is available; until then, the six-client federation remains a simulation of data locality, not a replication of it.

Algorithm coverage. The MLP path is now evaluated on all three substrates, and the LSTM path—which consumes the raw 60×24 windows and matches the temporal architectures that lead centralised SWAN-SF benchmarks [16]—has been evaluated federated on the raw substrate (Section 6.3, owner GPU, two seeds): the sequence arms do not collapse at natural prevalence, at window level and at event level, retroactively validating them as the route to closing the tree-ensemble gap in federated settings. The SCAFFOLD-LSTM arm has since been executed (0.970/0.294 window-level; 55/65 events at 0.41 false-alarm windows per day), recovering the hardest raw-substrate MLP failure, and a second-seed replication confirms every ranking result within ± 0.005 ROC-AUC (Section 6.4). What remains on this axis: their cleaned-fold pairing. Personalised FL (per-client heads over a shared backbone) is a further direction: agencies may prefer a global model for situational awareness plus a locally adapted model for their own asset fleet.

Event-level evaluation. All headline metrics in this paper are *window-level*: every 60-minute sliding window is scored independently, yet windows overlapping the same active region are strongly correlated, and one physical flare event triggers many correlated decisions. A 2% window-level FPR therefore does not mean 2% of operational alerts are false, and a detected event can generate a burst of duplicate alerts. The event-level evaluation module (detection

rate, missed events, alerts per detected event, duplicate-alert rate with cooldown deduplication, warning lead time, alert rate per day) is implemented, unit-tested, and documented for CLI use; executing it on the trained models requires event grouping keys and timestamps, which—like region identifiers—must be recovered from the raw SWAN-SF metadata. Until then, all recall and FPR figures in this paper carry the window-semantics caveat stated in Table 11’s caption.

Benchmark-construction effects and the raw-benchmark experiment. The cleaned variant’s training partitions are pre-balanced (RUS, Tomek links, TimeGAN augmentation), which (i) renders the SMOTE balancing arm inert—reported as a null result—and (ii) creates the $26\times$ prevalence shift that drives the calibration story. Because the cleaned variant bundles several interventions (imbalance handling, imputation, normalisation, synthetic augmentation), the in-partition results answered the question “can a federation learn the cleaned benchmark?” rather than “can a federation learn the underlying problem?”. This extension has now been *executed* (Section 6.2): the frozen protocol was retrained on the raw, unbalanced partitions with the release’s imputation and normalisation stages reproduced in-pipeline from their published description and every parameter fitted on the training pool only, with the reproduction verified against the released export on the audit-matched windows (median Spearman $\rho = 0.898$ at observed-nonzero positions). The outcome reassigns the federated versus centralised question entirely: the centralised arms are substrate-robust (logistic regression unchanged to three decimals; centralised MLP PR-AUC $0.382 \rightarrow 0.445$), XGBoost’s minority-class precision was inflated by the synthetic positives (PR-AUC $0.457 \rightarrow 0.329$), and every federated *MLP* arm degrades sharply on the natural-prevalence substrate (FedProx $0.976/0.308 \rightarrow 0.930/0.149$; FedAvg PR-AUC 0.056 and zero events detected; SCAFFOLD, executed here for the first time on a real substrate, fails at $0.768/0.057$)—while the sequence arms of Section 6.3 substantially restore the federated standing (FedProx-LSTM $0.970/0.404$), relocating the fragility from the federation mechanism to the flattened-feature representation. What remains open is fold-replication breadth: the seed-replication and natural-prevalence balancing questions queued in earlier revisions are now executed (Section 6.4: every arm’s ROC within ± 0.005 at seed 43; per-client SMOTE is a clean negative—FedAvg unchanged, FedProx -0.116 PR-AUC—so the rescue is architectural rather than class-balance-driven).

Executed and queued re-run programme. For auditability we collect the re-run experiments in one place. Five items of the original programme have since been *executed* in the independent re-execution (Section 5.6) and are reported in Section 6: the validation-frozen FPR operating-point report (realised test behaviour, quantified in Section 6.9); the calibration-comparison harness across raw / prior-shift / Platt / isotonic / temperature arms (Table 12); the untouched-holdout client evaluation (Section 6.12); the budget-matched centralised retraining at a 500-epoch cap (Section 5.5: the shipped reference was undertrained—early stopping fires at epoch 13—and at full budget parity the centralised MLP reaches $0.917/0.219$ versus FedProx’s $0.954/0.307$, so the headline gap survives budget parity); and the $\alpha=5.0$ promotion run at the full 50-round protocol (negative: $0.955/0.296$ versus $0.954/0.307$; $\alpha=1.0$ retained, with the incidental finding that plain FedAvg completes without collapse at $\alpha=5.0$, isolating the headline divergence as a heterogeneity-severity effect). The raw-metadata provenance audit subsequently resolved the metadata blockers and executed items (1)–(3) on the leakage-free fold (Section 6.1: region-disjointness holds by construction of the temporally-preceding fold; natural-prevalence test boundary; event-level evaluation with 65/65 GOES peak matches), and the raw-substrate retraining executed item (5) together with the SCAFFOLD half of item (4) on the natural-prevalence benchmark (Section 6.2), and the LSTM half of item (4) was executed on the owner GPU (Section 6.3: the sequence arms restore the federated standing, FedProx-LSTM $0.970/0.404$, at 2.6 h wall-clock against an estimated 74 h of CPU), and the second owner-GPU batch executed the remaining queue (Section 6.4: SCAFFOLD-LSTM, the seed-43 replication

of all four arms, and the natural-prevalence SMOTE ablation, 8.7 h, validated by a 64-check artefact audit). What remains queued is fold-replication breadth: the remaining temporally-preceding folds for the leakage-free and raw-substrate runs, and the sequence arms’ cleaned-fold pairing. Items previously labelled operational preconditions have been discharged; the event-level figures of Tables 5 and 7 are the operational performance figures of this revision, with the fold-replication caveat stated above.

9 Conclusion

We presented what the documented literature search of Section 2.4 indicates is the first application of federated learning to solar flare prediction, motivated by a plausible deployment constraint in operational space weather: several public data custodians restrict redistribution of raw magnetogram time series, so the strongest pooled models may not be trainable at all unless the custodians federate. The proposed framework simulates that constraint on the public SWAN-SF benchmark—six regional clients (neutral labels, no real institution) holding disjoint Dirichlet-skewed shards, MLP clients trained under FedAvg and FedProx with a clamped server-aware focal loss, and a frozen, leakage-audited evaluation protocol whose every selection decision is made on validation data. The study is an upstream flare-probability component for grid-resilience pipelines; the downstream flare→CME→storm →GIC chain is discussed as future integration, not evaluated.

To keep the summary honest, we classify the paper’s claims into three levels, tiered against the leakage-free fold of Section 6.1. **Demonstrated (leakage-free, single-pass, frozen protocol)**: the cleaned SWAN-SF export’s same-partition train/test pairing shares instances (100% of flaring test windows; 85.7–90% TimeGAN-synthetic training positives), and the benchmark’s intended temporally-preceding protocol removes the overlap; on that protocol all arms generalise (ROC-AUC 0.906–0.978); the in-partition federated-versus-centralised gap does not survive (FedProx 0.976/0.308 vs architecture-matched centralised MLP 0.973/0.382); FedProx retains checkpoint-independent ROC stabilisation over plain FedAvg (0.976 vs 0.906); the threshold-transfer and calibration failures persist (only XGBoost’s posteriors are usable); and event-level detection is essentially solved for every arm (64–65 of 65 events, 23.4 h median lead) while the false-alarm burden separates the arms by roughly 7×. Retraining the identical frozen protocol on the RAW, unbalanced benchmark (natural 2.05% training prevalence, no rebalancing, no synthetic positives, the release’s imputation and normalisation reproduced in-pipeline with train-only parameters and verified against the released export on the audit-matched windows) then separates the arms decisively: the centralised models are substrate-robust (logistic regression unchanged at 0.978; centralised MLP PR-AUC rises to 0.445), XGBoost’s minority-class precision was inflated by the synthetic positives (PR-AUC 0.457→0.329), and every federated *MLP* arm degrades sharply (FedProx 0.930/0.149; FedAvg detects zero of 65 events; SCAFFOLD, executed here for the first time on a real substrate, fails at 0.768/0.057, while the same algorithm’s LSTM instance recovers to 0.970/0.294). The sequence encoder qualifies the ending: the identical frozen protocol with the SolarLSTM, executed on the owner GPU, restores the federated arms substantially (FedAvg-LSTM 0.958/0.305 where the FedAvg MLP fell to 0.875/0.056; FedProx-LSTM 0.970/0.404, above its own pooled comparator 0.962/0.314 and XGBoost’s raw PR-AUC, though still below logistic regression 0.978/0.448, and at event level the substrate’s highest detection rate, SCAFFOLD-LSTM’s 55/65 at 0.4 false-alarm windows per day, bought at a higher alarm budget)—so the balanced-substrate standing is a property of the flattened-feature MLP arms, not of federation per se, and no federated arm exceeds the best centralised baseline on any substrate’s window-level ranking (the SCAFFOLD-LSTM event-level detection rate is the one priced exception, achieved at roughly twice the centralised arms’ false-alarm burden). **Demonstrated in-partition only (protocol-stability evidence, disclosed overlap)**:

FedProx with $\mu=0.01$ is the most *consistent* stabiliser among the components studied—above FedAvg on PR-AUC on five of five seeds (one-sided Wilcoxon $p = 0.031$) with $2.7\times$ narrower ROC-AUC variance, where the leakage-free fold shows the PR ordering between the two FL arms reverses; the FedProx global model attains 97.7% of the centralised XGBoost reference’s in-partition ROC-AUC (0.954 vs 0.977); plain FedAvg diverges under the preregistered configuration (and completes without collapse at $\alpha=5.0$, so the divergence is a heterogeneity-severity effect); per-client SMOTE is inactive on this benchmark variant (zero synthetic samples); and both the centralised and federated models concentrate attribution on current helicity, flux-emergence proxies, and Lorentz-force terms. **Supported with caveats:** one FedAvg cell with distribution-aware aggregation exceeds the FedProx headline ROC-AUC, recall-at-FPR figures are window-level statistics over correlated windows, five-seed confidence intervals quantify seed variability only, and the leakage-free fold is a single fold and seed (fold- and seed-replication is queued owner-side compute). Two formerly listed caveats were resolved by executed re-runs: the optimisation-budget confound (the budget-matched 500-epoch re-run of Section 5.5 shows the in-partition FedProx gap survives parity, though the gap itself is an in-partition artifact) and the within-distribution client holdouts (the untouched-holdout evaluation of Section 6.12 shows the federation benefit is larger under clean estimates). **Not demonstrated:** cryptographic privacy (secure aggregation and differential privacy are implemented but their accuracy cost is unmeasured; poisoning defence is absent), leakage-free multi-seed/ablation replication and the remaining temporally-preceding folds (one fold executed here), deployment-grade calibration (under the prior shift the neural posteriors are worse than climatology on Brier and ECE, and the executed five-arm comparison shows no validation-fit calibrator rescues them), neural-arm false-alarm reduction at event level, the downstream GIC/SCADA pipeline, the cleaned-fold pairing of the sequence arms, and fold-replication breadth; the SCAFFOLD-LSTM arm, the seed-43 replication, and the natural-prevalence balancing ablation queued in earlier revisions have since been executed (Section 6.4: the four-arm federation matrix is complete, every ranking result replicates within ± 0.005 ROC-AUC, and per-client SMOTE is a clean negative). An independent re-execution from the public, byte-verified dataset artefacts reproduces every in-partition headline metric exactly (Section 5.6); eight items of the re-run programme—the frozen-FPR report, the calibration arms, the untouched holdouts, the budget-matched and $\alpha=5.0$ re-runs, the raw-metadata provenance audit, the event-level/leakage-free evaluation on the temporally-preceding fold, and the raw-substrate retraining with the SCAFFOLD arm, plus the federated LSTM arms, their seed-43 replication, and the natural-prevalence SMOTE ablation—have since been executed with results in Section 6; the remaining items are listed in Section 8, each with the implemented-and-tested protocol that would upgrade the claim.

For the smart-grid community the message is correspondingly conditional: *if* custodians of magnetogram data face data-locality constraints, then at benchmark scale a proximal-regularised federation keeps the federation converging where plain averaging collapses, at negligible bandwidth, without any raw data movement—while matching the same-architecture pooled model’s ranking is, on the leakage-free protocol, a property the pooled reference shares rather than a federation advantage, and on the raw unbalanced benchmark that parity is encoder-dependent: with the flattened-feature MLP every federated configuration is strictly dominated by simple centralised baselines, while with the sequence encoder the best federated arm (FedProx-LSTM, 0.970/0.404) comes within 0.008 ROC-AUC and 0.044 PR-AUC of the best centralised baseline and is the first federated result whose Brier beats the climatology floor. No federated arm exceeds the best centralised baseline on any substrate, however, so the case for federating this problem still rests entirely on the data-locality constraint, not on predictive performance. For the machine-learning community the study is a caution and a demonstration: instance-level provenance audits can invert the conclusions drawn from a widely used preprocessed benchmark, and the cross-silo regime of the physical sciences has failure modes—prior mismatch, layered

imbalance handling, optimiser–variance-reduction interactions—that the medical and mobile FL literature does not fully anticipate, and that are worth studying in their own right.

A Configuration Evolution of the Released Framework

The released framework reached its present form through six documented version iterations (v2.1–v2.6), each responding to a diagnosed failure mode. We record this evolution because reproducibility of *process*—not only of final numbers—is what future deployments in other domains will actually need, and because several of the fixes correct subtle interactions that a clean-slate reimplementaion would likely rediscover the hard way.

Table 13: Documented stability fixes and protocol revisions across framework versions (v2.1–v3.8), as recorded in the released source.

Version	Diagnosed failure mode	Fix
v2.1	LSTM models require 3D input while baselines require 2D; a single data path served neither correctly.	Dual data path: 2D statistical features for MLP and baselines; raw 60×24 tensors for LSTM.
v2.2	F_β -optimal thresholds were computed but not applied; tables and confusion matrices reflected the default threshold.	Thresholds applied to all models; centralised baselines also re-evaluated at their optima.
v2.3	SCAFFOLD NaN every round: <code>std()</code> with Bessel correction on size-1 tensors divides by zero, poisoning control variates; DA-FL cap of $1/N$ starved large clients; focal α drifted to 0.95.	Population std for variate clipping; DA-FL cap $2/N$; alpha clamp tightened.
v2.4	Dynamic alpha scaling (up to $3\times$) re-inflated positive-class weight, undoing the focal- α fix; clients over-predicted flares (recall ≈ 1.0 , precision ≈ 0.03).	Square-root-damped imbalance factor capped at $1.5\times$; progressive factor narrowed to 0.9–1.1.
v2.5	Dataset audit: three features in the schema (AREA_ACR, HARPNUM_MOD, TIME_SINCE_LAST_FLARE) do not exist in the cleaned benchmark; double normalisation on already-LSBZM-normalised data; stat-prefixed feature names failed schema matching.	Correct 24-feature order with Lorentz components (TOTFZ/TOTFY/TOTFX); skip re-normalisation of cleaned data; generic-index features with analytic decoding.
v2.6	$\alpha=0.5$ partitions produced 3%–99% local flare rates, destabilising aggregation; 3 local epochs starved LSTM temporal learning; mixup on 3D sequences created unrealistic interpolations.	Dirichlet $\alpha \rightarrow 1.0$; local epochs $3 \rightarrow 10$; learning rate $10^{-3} \rightarrow 5 \times 10^{-4}$; mixup and SCAFFOLD disabled pending rework.
v3.0	Evaluation-protocol audit: thresholds selected on the test set; stale accuracy reported for federated models; non-disjoint Dirichlet shards; SMOTE dead code; DA-FL applied to both algorithm arms; one configuration constant scattered across modules.	Frozen protocol: validation-only selection, single-pass test, runtime leakage audit, disjoint-by-construction partitioning; explicit ablation switches (aggregation, SMOTE, loss, calibration); single <code>config.py</code> source of truth; centralised-MLP architecture control added.

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Version	Diagnosed failure mode	Fix
v3.0.1	Single-seed headline lacked statistical support; results not machine-readable end-to-end.	Frozen-protocol rerun on the full cleaned benchmark (seed 42, 50 rounds); five-seed study with Wilcoxon tests; 24-cell ablation grid; μ/α sweep; SMOTE-validity, interpretability, and communication artefacts; SHA-256 data manifest; 74-check integrity suite.
v3.1	External review: region overlap across train/validation; balanced-prevalence validation cannot support threshold transfer or calibration; client-level evaluation optimistic; centralized-FL optimisation budgets unstated; window-level metrics read as event metrics; client labels implied institutional participation.	Protocol extensions, all unit-tested (78 checks) and queued for the full-data re-run: region-disjoint splits, natural-prevalence validation with validation-frozen FPR operating points, untouched client holdouts, training-budget audit, event-level metrics, calibration-comparison harness; neutral client labels; threat-model table; documented literature-search protocol.
v3.2	Second external review (25 findings): scope/title mismatch, premise overreach, stale v2.x text, claim taxonomy, budget confound, calibration honesty, communication-cadence ambiguity.	Data-locality scope title; claims taxonomy (verified / benchmark-only / not demonstrated); threat-model and budget tables; ECE column; literature-search protocol; lit-search artefacts; three regenerated neutral figures; response map docs/REVIEW2_RESPONSE.md.
v3.3	Reproducibility unproven end-to-end; calibration arms, frozen-FPR realisation, and untouched holdouts still unexecuted on real data.	Independent re-execution from the public Cleaned SWAN-SF release (SHA-256-verified against the data manifest): every headline test metric of all six models reproduced exactly (bit-stable CPU protocol); calibration-arm comparison executed on the trained models (negative result: no validation-fit arm survives the $26\times$ prior shift); untouched-holdout client evaluation executed (local 0.983 / FedAvg 0.866 / FedProx 0.988 mean PR-AUC); frozen-FPR realised test FPR reported; budget-matched 500-epoch centralised re-run executed (the shipped reference was undertrained; at parity 0.917/0.219 vs FedProx 0.954/0.307—gap survives); $\alpha=5.0$ promotion executed (negative; $\alpha=1.0$ retained); fig:clients regenerated from the clean protocol; runners made cache-reusing and round-resumable.

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Version	Diagnosed failure mode	Fix
v3.4	Same-partition protocol: train/test instance overlap unknown and assumed absent; event-level and leakage-free evaluation impossible without raw metadata; four re-run items queued on that dependency.	Raw SWAN-SF benchmark obtained (Harvard Dataverse) and every cleaned window aligned to its raw instance via normalization-invariant matching (98.7–100% verification, 100% label agreement): instance-sharing quantified (test exports contain all raw instances; 100% of flaring test windows are training windows; 85.7–90% of training positives TimeGAN-synthetic); leakage-free fold executed under the benchmark’s intended temporally-preceding protocol (train P1–4 → test P5, round-resumable runner <code>run_partition_disjoint.py</code>): central-vs-FL gap disappears (FedProx 0.976/0.308 vs centralised MLP 0.973/0.382), FedProx ROC stabilisation over FedAvg survives (0.976 vs 0.906, checkpoint-independent), PR ordering reverses (FedAvg 0.370 above FedProx 0.308); calibration/threshold-transfer failures persist; event-level evaluation executed on 65 physical M/X-class events (detection 98.5–100%, XGBoost false-alarm burden $\sim 7\times$ lower than neural arms, 23.4 h median lead to flare peak); claims re-tiered, in-partition results disclosed as such.
v3.5	Raw-substrate retraining (re-run item 9) and the SCAFFOLD arm (item 8, half) still queued; the release’s imputation/normalisation stages not reproducible in-pipeline.	Both executed (<code>experiments/raw_substrate.py</code> , <code>run_raw_substrate.py</code> , <code>run_event_level_raw.py</code>): FPCKNN-style imputation and LSBZM-style normalisation reproduced from the release paper’s description with train-only parameters and verified against the released export on the audit-matched windows (median Spearman $\rho = 0.898$ at observed-nonzero positions; release is an exact monotone image of raw, $\rho = 1.000$; R-value zero mass re-imputed by the release but preserved here); frozen protocol retrained on raw P1–4 at natural 2.05% prevalence, single-pass test on raw P5: centralised arms substrate-robust (LR 0.978/0.448, XGB 0.974/0.329, MLP 0.971/0.445), federated arms collapse (FedAvg 0.875/0.056, 0/65 events; FedProx 0.930/0.149; SCAFFOLD 0.768/0.057); event-level: central arms 61.5–80% detection at 0.2–0.3 FA windows/day; claims re-tiered across three substrates.

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Version	Diagnosed failure mode	Fix
v3.6	Federated LSTM arm (item 8, second half) unexecuted on owner-side compute; raw-substrate collapse established only for the MLP arms, leaving its architecture dependence open.	LSTM arms executed on the owner GPU (RTX 4060, 2.6 h; <code>run_federated_lstm.py</code> , <code>outputs/raw_lstm_eval.json</code> ; substrate identity asserted): centrally the LSTM is the weaker architecture (0.962/0.314 vs 0.971/0.445) but the federated sequence arms do not collapse—FedAvg-LSTM 0.958/0.305, FedProx-LSTM 0.970/0.404, above its pooled comparator and XGBoost’s raw PR-AUC, below logistic regression; first federated Brier to beat the climatology floor; stabilisation now on three substrates $\times$ two encoders; CPU-only event-level runner released; claims re-tiered (the collapse is a flattened-feature-MLP property).
v3.7	Event-level behaviour of the LSTM arms unevaluated: the owner-GPU run predated the audit-metadata extraction, so the v3.6 release shipped the CPU runner staged but unexecuted against real probabilities.	Executed via <code>run_event_level_lstm.py</code> (torch-free, CPU, Windows <code>_aux</code> fallback added) over the GPU run’s stored test probabilities and the audit metadata (65/65 GOES peaks matched, labels asserted equal): FedAvg-LSTM 35/65 events (53.8%, 0.2 FA windows/day) against its MLP counterpart’s 0/65; FedProx-LSTM 49/65 (75.4%) at 0.4 FA windows/day with a 23.2 h median lead—FedProx-MLP’s detection at $3.5\times$ lower false-alarm burden; no federated sequence arm reaches the 78.5–80% centralised detection frontier. Artefact <code>outputs/event_level_raw_lstm_p5.json</code> .
v3.8	The remaining owner-side programme unexecuted: SCAFFOLD-LSTM absent (hardest MLP failure left open), sequence arms single-seed single-device, and the natural-prevalence SMOTE ablation unstudied.	Executed as one resumable owner-GPU queue (8.7 h, RTX 4060; queue driver <code>run_gpu_queue.py</code> ; artefacts validated on receipt by a 64-check audit: protocol fingerprints, substrate identity, embedded frozen-counterpart bit-matches, event-key reconciliation, log cross-checks): SCAFFOLD-LSTM 0.970/0.294 window-level (vs MLP counterpart 0.768/0.057) and 55/65 events (84.6%) at 0.41 FA windows/day—the substrate’s highest detection rate, above the centralised frontier at a priced alarm budget; seed-43 replication of all four arms (ROC within ± 0.005 on every arm, qualitative structure unchanged); per-client SMOTE ablation a clean negative (FedAvg 0.961/0.307 unchanged, FedProx 0.946/0.288, -0.116 PR-AUC). Artefacts <code>raw_lstm_scaffold.json</code> , <code>raw_lstm_seed43.json</code> , <code>raw_lstm_smote.json</code> and their event-level JSONs; Tables 8–9 extended.

Three of these fixes deserve narrative emphasis because they encode non-obvious engineering knowledge. The SCAFFOLD failure (v2.3) is the most instructive: the NaN symptoms pointed nowhere near their cause, which was a statistics API detail—Bessel’s correction on single-element bias tensors—interacting with an algorithm whose correctness depends on exact control-variate arithmetic under fixed learning rates (AdamW’s adaptive steps further invalidate the update formula). The focal-alpha saga (v2.3–v2.4) illustrates layered imbalance handling drifting into degeneracy: SMOTE, class-weighted loss, and rate-aware aggregation all push the same effective prior, and unbounded adaptation at any one layer can silently dominate the others. The dataset audit (v2.5) is a reminder that in applied ML the schema *is* the experiment: three phantom features had been silently participating in the pipeline until an attribute-by-attribute audit against the benchmark’s actual storage order replaced them with the physically meaningful Lorentz-force components. The v3.x revisions add a fourth, blander lesson: protocol defects (test-set selection, optimistic client holdouts, unstated budgets) do not announce themselves in any loss curve—only an external audit found them—which is the argument for freezing the evaluation contract in code, as the released framework now does. The published artefacts, figures, and this paper’s numbers all post-date these fixes.

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